

- 1. MINERAL NUTRIENT DISORDERS AND
CONTROL**
- 2. CITRUS DISEASES AND CONTROL**
- 3. CITRUS INSECTS AND CONTROL**

CITRUS VARIETIES OF PAKISTAN

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Islamabad**

Citrus varieties

Preamble

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FOREWORD

Consequent to the implementation of the Seed Act, 1976, the seed industry project was launched the same year. The infrastructure established both at provincial and federal levels has shown excellent performance for providing significant quantum of quality seed of a number of field crops and remarkably successful in creating awareness in the farming community about the use and merits of quality seed. To meet the increasing demands of quality seed, the Government encouraged participation of private sector and consequently, a large number of seed companies entered seed industry.

The successful implementation of the quality control legislation has helped improve the availability of quality seed of field crops. However, little attention had been paid to the vegetables whereas the fruit crops had totally been ignored. It is highly appreciated that the Project for Horticultural Promotion (PHP), NWFP took the initiative and started producing and marketing of quality seed of various vegetables in collaboration with the public sector agricultural research institutions and private sector farming community. The training and awareness created at the grass root level through farm field schools has proved very useful in achieving the objectives.

Simultaneously, the PHP took notice of the nursery trees provided through ordinary nurseries to the farmers which generally were unhealthy, unpredictable and of poor quality. The efforts of the PHP and the Federal Seed Certification Department (FSC&RD) are highly appreciated particularly regarding the selection of genuine fruit trees; determine morphological description, and above all planting the representative fruit trees in the Germplasm Units (GPUs). The GPUs will usher a new era by providing true to varieties healthy propagating materials for producing and marketing vigorous and reliable fruit nursery trees by joint efforts of the government and private sector nursery men.

For determining genuineness and production of true to type nursery trees the basic requirement is the availability of accurate description of plant and fruit characteristics of mother trees and registered seed and scion stocks. Further to regulate the commercial production of nursery trees, the most important step is the registration of private sector nurseries. It is expected that with the implementation of the proposed nursery registration program and nursery tree certification standards will augment availability of quality nursery trees to the farmers. The deterioration which generally accrues due to non-availability of certified nursery trees will be held back. It will not only increase the yield of the

farm but also improve the uniformity and quality for local and export markets.

The Federal Seed Certification and Registration Department has accomplished the most basic and important task of establishing descriptions of citrus varieties. The descriptions given this book “**Citrus Varieties of Pakistan**” are expected to go a long way in improving the quality of the propagating materials and maintenance of true to type representatives of our fruit assets. The book will further provide officially documented information on the invaluable wealth of horticultural part of our biological diversity.

The book contains almost all the citrus varieties that have been developed and propagated for commercial production in the country. In addition to the descriptions, the book contains some background information on the parentage and origin of the variety, the breeder and the breeding centre along with the status of the variety.

Hopefully, the information presented in this publication will prove highly useful for the horticulturists, seed quality controllers, planners, nursery men and farmers. Earlier, the Department has prepared similar books on Wheat, Barley, Cotton and Mango Varieties of Pakistan and I wish the Department to arrange further publications of similar nature on other field and fruit crops.

SECRETARY

MINISTRY OF FOOD, AGRICULTURE AND LIVESTOCK

INTRODUCTION

Citrus fruits are the largest fruit crop in the world grown all around in over 140 countries. Major citrus growing countries include Brazil, the Mediterranean countries, USA and China. These countries represent more than two thirds of global citrus fruit production.

SUMMARY

	Variety	Origin	Fruit color	Size	Flesh color	juice %	Juice color	Acidity%	TSS %	Seeds
Mandarins, Tangerines										
1.	Dancy	Cal	RE	Med	DEO	41	W	0.9	8-9	many
2.	Fairchild		DEO	Med	OY	40	W	0.9	10-12	many
3.	Fewtrell	Aus	OR	Med	OD	36	W	0.4	9-11	many
4.	Fremont	Cal	BR	Med	DER	47	W	0.7	10-12	many
5.	Honey Mandarin	Cal	YG	Med	CRE	39	CRE	0.5	8-9	many
6.	Kino	Flo	OY	Med	DEY	47	LPIN	0.7	12-14	many
7.	Minneola	Flo	REO	Med	RO	50	W	0.6	8-9	Vs
8.	Orlando	Cal	OR	MI	OY	40	W	0.3	10-11	many
9.	Pearl	Cal	LY	Med	LO	36	W	0.7	9-10	Many
Sweet Oranges										
10.	Ari Pride	Loc	OY	Med	W	38	W	0.4	10-11	Few
11.	Blood Red	Cal	OR	Med	OY	42	LG	0.5	10-11	Med
12.	Campbell Valencia	Cal	OY	MI	OY	43	LG	0.9	10-11	Vs
13.	Casagrande	Cal	OY	Med	OW	43	W	0.4	9-10	Few
14.	Fisher Navel	Cal	DEO	MI	LY	39	LCRE	0.4	9-10	Vs
15.	Frost Valencia	Cal	OY	MI	OY	44	LG	0.7	9-11	Vs
16.	Jaffa	US	OY	Med	LO	42	W	0.5	9-11	Many
17.	Lanelate Valencica	Cal	OY	MI	LY	39	LY	0.4	9-10	Med
18.	Marrs Early	Cal	GY	Med	CRE	36	CRE	0.3	10-12	Vs
19.	Morrow	Cal	OY	MI	DEP	34	PIN	0.5	8-9	Few
20.	Mosambi	Moz	LG	Med	PY	40	W	0.2	9-11	Many
21.	Olinda Valencia	Cal	OY	MI	OY	45	LG	0.9	10-11	Few
22.	Parson	Flo	OY	MI	DUO	39	LCRE	39	9-10	Med
23.	Pineapple	Cal	GLO	Med	GOY	42	W	0.5	10-11	Many

Citrus varieties

Preamble

	Variety	Origin	Fruit color	Size	Flesh color	juice %	Juice color	Acidity%	TSS %	Seeds
24.	Ruby Blood	Cal	DEO	Med	ORRS	42	LR	0.5	9-11	Few
25.	Salustiana	Spa	OY	Med	OW	42	W	0.4	10-12	Vs
26.	Sanguinelli	Ital	REGB	Med	RERD	36	RE	0.5	9-10	Vs
27.	Spring Navel	Cal	DEO	MI	CRY	39	LY	0.4	9-10	Vs
28.	Succari	Int	OY	Med	PY	36	W	0.18	9-10	Many
29.	Sugar Malta	Int	OY	Med	W	36	W	0.15	10-11	Many
30.	Tarocco	Ital	OYB	Med	OYPG	43	W	0.5	9-10	Vs
31.	Valencia Late	Cal	OY	Med	OY	43	LG	0.9	10-11	Vs
32.	Washington Navel	Cal	DEO	Med	LCRE	39	LCRE	0.4	9-10	Vs
Grapefruits										
33.	Buntan Pummalo	US	YG	VI	LPIN	34	LPIN	0.19	8-9	many
34.	Duncan	US	LY	Med	LY	36	LY	1.3	9-10	Few
35.	Flame	US	LYB	Med	R	41	PINK	1	7-8	Vs
36.	Frost Marsh	Cal	LY	MI	W	47	OW	1	8-9	Vs
37.	Marsh JB	Cal	LYB	Med	WRS	41	RE	1.5	7-8	Few
38.	Marsh Seedless	Flo	LY	Med	BUF	39	LY	1.4	9-11	Vs
39.	Pink Ruby	Fra	LYB	MI	RDEP	36	PIN	1	7-8	Vs
40.	Ray Ruby	Fra	LYB	MI	RDEP	36	PIN	1	7-9	Vs
41.	Redblush	Cal	YB	MI	R	36	RE	1	9-10	Vs
42.	Reed	Cal	LYB	MI	WCRE	41	CRE	1	8-10	Few
43.	Ruby Blood	Cal	LYB	MI	PIN	48	PIN	1.3	9-10	Vs
44.	Shamber	Cal	LYB	MI	R	48	RE	1.3	9-10	Vs
45.	Star Ruby	Cal	LYB	MI	PKY	41	DRE	1.5	9-11	Vs
46.	Webber Shaddock	US	LY	L	LCRE	med	LPIN	0.15	8-9	many
Limes and Lemons										
47.	Chinese Lime	Int	LG	Sma	GSH	46	W	3.5	5-6	many

Citrus varieties

Preamble

	Variety	Origin	Fruit color	Size	Flesh color	juice %	Juice color	Acidity%	TSS %	Seeds
48.	Deperse	Fra	LG	Med	LG	52	W	3.5	5-6	many
49.	Eurika Lemon	Cal	LG	MI	GY	51	W	5	7-8	Few
50.	Frost Lisbon	Fra	LG	Med	GY	46	W	4.5	7-9	Few
51.	Galgai	Loc	PY	MI	PY	34	W	5.5	7-8	many
52.	Kaghzi Lime	Loc	LGR	sma med	GSH	52	W	4	5-6	many
53.	Lisbon	Cal	LGR	Ned	GSH	49	W	4.5	7-8	Few
54.	Mexicane	Fra	LGR	sm med	LG	52	W	3.5	5-6	many
55.	Rough Lemon	Loc	LGR	Med	LY	36	W	4.5	4-5	many
56.	Sweet Lime	Loc	LGR	Med	STR Y	46	W	0,2	8-9	many
57.	Tahiti Lime	Cal	LGR	Med	PG	43	W	4.5	7-8	Vs

Kumquats

58.	Maiwa	Int	DAO	Small	CRE	hi	CRE	Low	-	Few
59.	Nagami	Int	DAO	Sma	CRY	hi	W	Low	-	Few

VS Virtually seedless

Note:

ORIGIN			
CAL	California	ITA	Italy
FLO	Florida	LOC	Local
FRA	France	MOZ	Mozambique
INT	Introduction		

COLOR			
BR	Bright red	ORRS	Orange with red streaks
CRE	Creamy	OR	Orange red
CRY	Creamy yellow	OYB	Orange yellow with red blush

DAO	Dark orange	OYPG	Orange yellow pigmented
DEO	Deep orange	PG	Pale green
DEP	Deep pink	PIN	Pinkish
DEY	Deep yellow	PKY	Pinkish yellow
DRE	Dark reddish	PY	Pale yellow
DUO	Dull orange	R	Red
GLO	Greenish light orange	RDEP	Reddish deep pink
GOY	Greenish orange yellow	RE	Reddish
GSH	Greenish	REGB	Reddish green with red blush
GY	Greenish yellow	REO	Reddish orange
LCRE	Light creamy	STRY	Straw yellow
LG	Light golden	W	Whitish
LPIN	Light pink	WCRE	Whitish creamy
LR	Light red	WRS	White with red streaks
LYB	Light yellow with red blush	YB	Yellow with red blush
OY	Orange yellow	YG	Yellowish green
OW	Off-white		
LY	Light yellow		

MANDARINS and TANGERINES

DANCY

Type: Tangerine (*Citrus reticulata*); Dancy Tangerine, attractive midseason, medium bearer variety.

Origin: Introduction from California 1973; Origin a chance seedling selection in Florida.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Height medium, upright, bushy, branches upright to semi spreading, water shoots about spineless, fruit mostly on outer side, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, dense, size medium, rolled inwards, leaf undulation weak, length 7-9, width 3-5, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Bright reddish orange, size medium, oblate, length 6-7, width 5-6 cm, base round, apex flat, circular ring occasional, faint, rind thin, medium tight, peel very easy, puffy, surface smooth, oil glands size bold, density sparse, prominent; albedo white, colomella medium, solid, segments 10-14, separating easily.

Flesh and Flavor: Deep orange yellow, fibers soft, vesicles, tender, melting, juice high 40-42%, whitish, acidity 0.8-1.0%, TSS 8-9%, aroma medium, taste sweet flavor excellent, sprightly, distinctive.

Seed: Many seeds 8-15, size medium, color white, polyembryonic, pointed, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Dancy: Introduction from Florida, medium bearer Tangerine, tree height medium, upright, foliage dark green, dense; leaf petiole wings absent, flowering once, fruit medium, oblate, bright reddish orange, rind thin, medium tight, peel very easy, puffy, oil glands bold, prominent, segments separating easily, flesh deep orange yellow, vesicles soft, tender, melting, juice high, taste very good, flavor excellent, aromatic.

FAIRCHILD

Type: Tangerine (*Citrus reticulata*); Early maturing, medium bearer commercial variety grown on small scale.

Origin: Introduction from California, 1973; Origin a hybrid (*Clementine mandarin x Orlando tangelo*).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, bushy, branches spreading with drooping tendency, water shoots slightly thorny, fruit hold on tree good, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, dense, leaf shape long, narrow, undulation weak, length 8-10, width 3-4 and petiole 1-2 cm, petiole wings absent, aroma strong, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Deep orange, medium, medium oblate, length 6-7, width 5-6, base round, apex flat, rind medium thin, easy peel, medium soft, smooth, oil glands medium, sparse, less prominent; albedo white, colomella medium, solid, segments 10-14, separating easily.

Flesh and Flavor: Orange yellow, fibers firm, vesicles medium tender, melting, juice high (38-41%), color whitish, acidity 0.8-0.9%, TSS 10-12%, separating easily, aroma medium, taste pleasant rich in flavor

Seed: Many seeds 5-12, size medium, color white, polyembryonic, pointed, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Fairchild: Introduction from California, originated as a cross between Clementine mandarin x Orlando Tangelo, medium bearer mandarin, maturity earlier than Kino, tree medium tall bushy spreading, foliage medium green, dense; petiole wings absent, flowering once, fruit medium, smaller than Kino, oblate, deep orange, rind medium thin, easy peel, surface smooth, segments separate easily, flesh orange yellow, many seeded, vesicles medium tender, juice high, color whitish, aroma medium, flavor rich, tasty.

FEWTRELL

Type: Mandarin (*Citrus reticulata*); *Fewtrell's Early*, *Fewtrell*, *Feutrell*, *Fruiter*, early season, high yielding, major commercial variety.

Origin: Introduced from Australia; Origin unknown.

Maintainer: ORI, Sargodha, GPU, Sherkhana.

Tree: Medium tall, upright, branches semi spreading but recurve under fruit load, water shoots medium thorny, moderate tendency for alternate bearing (Rootstalk Rough Lemon).

Leaf: Foliage dark green, dense, leaf undulation very weak, size medium, length 7-9, width 4-5 and petiole 1-1.5 cm, petiole wings narrow, aroma medium, refreshing.

Flower: Flowering season spring, flower single, anther yellow, aroma strong, pleasant.

Fruit: Orange red, size medium, sub-globose to broadly obovate, length 6-7, width 5-6, base round, apex flat with brownish circular mark, rind medium thick, medium loose, easy peel, surface finely pitted, medium smooth, oil glands size medium, density medium, medium conspicuous; albedo white, colomella medium, semi hollow, segments 10-14, easily separating.

Flesh and Flavor: Orange red, fibers weak, vesicles medium, melting, juice medium 35-38%, color whitish, acidity medium 0.35-0.40 TSS 9-11%, flavor fine, rich, tasty, aroma strong pleasant.

Seed: Many, 5-15, size medium, color white, polyembryonic, pointed, cotyledons light green, inner coat light brown.

Yield: Good bearer.

Leading Characters: Fewtrell: Introduction from Australia, medium yielder mandarin, tree medium tall, upright, foliage dark green, leaves smaller in size than Kino, flowering once, fruit size medium, smaller in size than Kino, orange red, surface finely pitted, rind medium thick, medium loose, easy peel, colomella semi hollow, flesh orange red, medium juicy, flavor fine rich aromatic.

FREMONT

Type: Tangerine (*Citrus reticulata*); attractive early maturing, medium bearer commercial variety.

Origin: Introduction from California 1973; Origin hybrid (*Clementine* x *Pounkan mandarin*).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Height medium, bushy, upright, branches semi spreading, water shoots spineless, fruit mostly on outer side, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium sparse, leaf size medium, moderately rolled, undulation weak, length 8-10, width 3-4 and petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Bright reddish orange, size medium, oblate, length 6-7, width 5-6 cm, base round, apex flat with circular ring on majority of fruits, rind thin, medium tight, peel medium, surface smooth, oil glands size bold, sparse, prominent; albedo white, colomella medium, solid, segments 10-14, separating easily.

Flesh and Flavor: Deep orange yellow, fibers soft, vesicles, tender, melting, juice high 45-50%, whitish, acidity 0.7-0.8%, TSS 10-12%, aroma medium, taste sweet flavor excellent, sprightly, distinctive.

Seed: Many 8-15, medium, white, polyembryonic about half mono-embryonic, pointed, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Fremont: Introduction from Florida, medium bearer hybrid Tangerine, tree height medium, bushy, foliage dark green, medium dense; leaf petiole wings absent, fruit size medium, oblate, color bright reddish orange, rind thin, medium tight, peel medium, surface smooth, oil glands bold, prominent, segments separating easily, flesh deep orange yellow, vesicles soft, tender, melting, juice high, color whitish, flavor rich excellent tasty, distinctive, sprightly.

HONEY MANDARIN

Type: Mandarin (*Citrus reticulata*); very early season, medium bearer variety of Mandarin.

Origin: Introduction, California 1973; Origin a hybrid sister line of Kino (King x Willowleaf).

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium tall, vigorous, upright, water shoots medium thorny, medium tendency for alternate bearing, (Rootstalk Rough Lemon),

Leaf: Foliage dark green, dense, leaf undulation weak, length 7-9, width 4-5, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering season spring, flower single, anther yellow, fragrance strong, pleasant.

Fruit: Yellowish green, size medium, oblate round, length 5-7, width 5-6 cm, base rounded slightly elevated with ridges, apex flat slightly depressed, rind medium thick, medium loose, easy peel, surface rough, faintly pebbled, oil glands size medium, density medium, prominent; albedo white, columella medium, semi solid, segments 10-14 separating easily.

Flesh and Flavor: Creamy, fibers weak, vesicles medium, melting, juice high 38-40%, juice color creamy orange, acidity 0.45-52%, TSS 8-9%, aroma strong, very pleasant, flavor fine, rich, sweet.

Seed: Many seeded 5-20, medium, white, polyembryonic, pointed, cotyledons light green, inner coat light brown.

Yield: medium to good cropper.

Leading Characters: Honey Mandarin: Introduction from California, hybrid sister line of Kino, medium bearer mandarin, tree medium tall, upright, foliage dark green, medium dense, petiole wings absent, flowering once, fruit size medium, oblate round, resembles Kino except elevated base, yellowish green, rind medium thick, medium loose, medium rough, oil glands size medium, density medium, conspicuous, flesh creamy pink, melting, juicy, flavor rich, aromatic.

KINO

Type: Mandarin (*Citrus reticulata*); Kinnow, Kino, midseason, most popular high bearer commercial variety.

Origin: Introduction from California, 1940; Origin a hybrid, (King x Willowleaf Mandarin).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, wide spreading, branches slender, long, semi spreading, water shoots weak thorny, alternate bearer, a large crop of smaller fruits followed by a small crop of larger fruits, fruit hold on tree and storage very good (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, length 8-10, width 4-5, petiole 1-2 cm, petiole wings absent, aroma refreshing.

Flower: Flowering spring, solitary, anther yellow, fragrance strong.

Fruit: Orange-yellow, size medium, oblate round, length 6-8, width 7-9 cm, base and apex flat, slight to medium depressed, rind medium thin, medium loose, easy peel, slightly pitted, surface smooth, glossy, oil glands size medium, density medium, prominent; albedo white, colomella medium, semi hollow, segments 10-14, separating easily.

Flesh and Flavor: Deep yellow orange, fibers weak, vesicles medium, quick melting, juice high 46-48%, color light pink, acidity 0.6-0.8%, TSS 12-14%, aroma strong and distinctive, flavor rich and wonderful.

Seed: Many seeded 5-30, size medium, white, polyembryonic, pointed cotyledons light green, inner coat light brown.

Yield: High bearer, highest harvest 5300/tree at Sargodha.

Leading Characters: Kino: Introduction from Florida, most popular high bearer midseason mandarin, tree medium tall, vigorous, foliage dark green, medium dense; petiole wings absent, flowering spring, fruit size medium, oblate round, orange yellow, rind medium thin, medium loose, flesh deep yellow orange, melting, juice high, flavor excellent, sweet, aromatic.

MINNEOLA

Type: Tangelo (*Citrus reticulata*); medium late maturing, medium bearer commercial variety grown on small scale.

Origin: Introduction from Florida, 1973; Origin hybrid sister line of Orlando (Duncan Grapefruit x Dancy Tangerine).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium to tall growing, vigorous, bushy, branches spreading, water shoots spineless, regular bearer (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf long, broad, pointed, leaf undulation weak, length 9-11, width 5-7, petiole 1-2 cm, petiole wings small, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, fragrance medium, cross pollination recommended for regular, good bearing, cross-incompatible.

Fruit: Deep reddish orange, size medium to large, oblate to bell shaped (Sangtra type), neck long, length 7-8, width 8-9 cm, base round, rind medium thick, medium loose, easy peel, surface smooth, oil glands size medium, sparse, less prominent; albedo white, colomella medium, hollow, segments 10-14, loose, separating easily.

Flesh and Flavor: Reddish orange, fibers soft, vesicles fender, juice high 50-55%, color whitish, acidity low 0.6-0.7%, TSS 8-9%, flavor rich, aroma medium, pleasant.

Seed: Virtually seedless 3-6, size medium, white, polyembryonic, pointed cotyledons light green, inner coat light brown.

Yield: Medium to good bearer.

Leading Characters: **Minneola**, Introduction from Florida, hybrid, wonderfully combines the sweetness of a Mandarin with the tart bite of a Grapefruit, attractive color, excellent flavor, low seed contents, tree medium to tall bushy, foliage dark green, fruit large, neck long, deep reddish orange, rind medium thick, flesh reddish orange, vesicles soft, juice high, flavor excellent, identification very easy, Knob on top.

ORLANDO

Type: Tangelo (*Citrus reticulata*); medium early maturing, medium bearer variety of Tangelo.

Origin: Introduction from California 1973; Origin hybrid, a sister line of Minneola, (Duncan Grapefruit x Dancy Tangerine).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, bushy, moderately vigorous, branches spreading, water shoots spineless, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf size medium, strongly rolled, leaf undulation medium, length 7-9, width 4-5, petiole 0.5-1.0 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, fragrance medium, pleasant, cross-pollination recommended for regular and good bearing.

Fruit: Orange red, size medium large, oblate to sub-globose, length 6-7, width 7-8 cm, base round, apex flat, rind thin, medium tight, easy peel, surface smooth, slightly pebbled, oil glands size small, numerous, less prominent; areola medium, albedo white, colomella medium, hollow, segments 12-14, separating easily.

Flesh and Flavor: Orange yellow, fibers soft, vesicles medium tender, juice high 38-42%, color whitish, acidity low 0.3-0.4%, TSS 10-11%, aroma medium, flavor pleasant, mildly sweet, aromatic.

Seed: Many seeded 5-10, size medium, white, polyembryonic, pointed, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Orlando: Introduction from California, hybrid sister line of Minneola, tree medium tall bushy spreading, requires cross-pollination for regular and good fruiting, foliage medium green, medium dense; fruit medium large, oblate to sub-globose, orange red, rind thin, medium tight, smooth slightly pebbled, flesh orange yellow, flavor rich, milder and slightly sweeter than Minneola.

PEARL

Type: Tangelo (*Citrus reticulata*); medium early maturing, medium bearer variety grown on small scale, hold on tree affect quality.

Origin: Introduction from California 1973; Origin a hybrid, (Imperial Grapefruit x Willowleaf Mandarin).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, Vigorous, bushy, branches droopy, water shoots almost spineless, alternate bearer, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation absent, medium, length 4-6, width 3-5, petiole 1-1.5 cm, petiole wings very narrow, aroma medium, refreshing.

Flower: Flowering throughout, mainly spring, aroma strong.

Fruit: Light yellow, size medium to small, slightly oblate, length 6-7, width 5-6 cm, base round, apex flat with slight to medium naval, areola medium, rind thin, medium tight, easy peel, surface finely pitted, medium smooth, oil glands size medium, density medium, medium conspicuous; albedo white, colomella medium, solid, segments 10-14 separating easily.

Flesh and Flavor: Light orange, fibers weak, vesicles tender, juice medium 35-38%, color whitish, acidity medium 0.7-0.8%, TSS 9-10%, fine rich flavor pleasantly sweet, aroma strong, pleasant.

Seed: Many seeded 20-30, size medium large, white, polyembryonic, pointed, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Pearl: Introduction from California, medium cropper hybrid Tangelo, tree medium tall, foliage dark green, flowering throughout the year, petiole wings very narrow, fruit medium, slightly oblate, apex flat with slight to medium naval, color light yellow, rind thin, medium tight, easy peel, oil glands medium, medium dense, medium conspicuous, seeds many, flesh color light orange, medium juicy, whitish, flavor fine, rich, tasty, aromatic.

Citrus varieties

Sweet Oranges

SWEET ORANGES

ARI PRIDE

Type: Sweet Orange (*Citrus sinensis*); medium early maturing, medium bearer variety.

Origin: Local hybrid, Pineapple x Mosambi

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, bushy, medium vigorous, branches semi spreading, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf size medium, broad, undulation weak, length 7-9, width 3-5, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Orange yellow, size medium, roundish, length 6-9, width 7-8 cm, base round, apex flat with noticeable circular ring like Mosambi, rind medium thin, medium loose, surface smooth, oil glands small, numerous, medium conspicuous; albedo white, colomella medium, solid, segments 10-14.

Flesh and Flavor: Whitish, fibers soft, vesicles medium melting, juice medium 38-40%, juice color whitish, acidity low 0.40-48%, TSS 10-11%, aroma medium, taste pleasant, aromatic.

Seed: Few seeded 7-10, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: ARI Pride: Local hybrid, medium bearer sweet orange, tree medium tall bushy, medium vigorous, branches semi spreading, medium spiny, foliage color medium green, medium dense, leaf medium broad, petiole wings absent, flowering once, fruit size medium, roundish, color orange yellow, apex flat with noticeable circular ring but less prominent than Mosambi, rind medium thin, medium loose, smooth, oil glands small, medium conspicuous, flesh whitish, vesicles soft, medium melting, juice medium (more than Mosambi), color whitish, flavor rich, taste very good, aromatic.

BLOOD RED

Type: Sweet Orange (*Citrus sinensis*); (Blood Red, *Red Blood Malta*) mid to late season, medium bearer commercial variety.

Origin: Introduction from California 1973; Origin Mediterranean.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium tall, vigorous, wide spreading, branches semi spreading but recurve under fruit load, water shoots medium thorny, fruit hold on capacity and storage very good (Rootstalk Rough Lemon).

Leaf: Foliage dark green, leaf undulation very weak, size bigger, broader, length 8-10, width 4-6, petiole 1-2 cm, petiole wings narrow, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Deep orange, red flushed on sides, size medium, globose to obovate, length 6-7, width 6-8 cm, base and apex flat, medium depressed, rind thin, tight, peel medium, finely pitted, medium smooth, oil glands size medium, density medium, less conspicuous; albedo white, colomella medium, solid, segments 10-14.

Flesh and Flavor: Orange yellow to deep red, fibers medium, vesicles medium, fine grained, juice high 40-45.0%, juice color light golden to red, acidity 0.5-0.6%, TSS 10-11%, fine rich flavor and aromatic.

Seed: Moderately seedy 8-16, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Blood Red: Introduction from California, mid to late maturing orange, tree medium tall, vigorous, foliage dark green, flowering once, fruit medium, globose to obovate, deep orange, red flushed on sides, rind thin, tight, flesh orange yellow to blood red, fine grained, good juicy, light golden to red, flavor rich, taste fine, refreshing. Development of color is highly variable and uncertain, color more pronounced, attractive with excellent flavor in fruits produced on trees in sub-mountainous regions.

CAMPBELL VALENCIA

Type: Sweet Orange (*Citrus sinensis*); Campbell, Campbell Valencia, a very late season high bearer variety of sweet orange.

Origin: Introduction from California, 1973; Origin most probably a chance seedling selection.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium short, vigorous, compact, branches semi spreading, water shoots more thorny, tends to be alternate bearing especially if the fruit is harvested very late in the season (Rootstalk **Rough Lemon**).

Leaf: Foliage medium green, medium dense, leaf undulation very weak, size large, length 8-10, width 4-6, petiole 1-2 cm, petiole wings very narrow, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Orange-yellow sometimes with greenish tinge, medium large, ellipsoid, length 7-9, width 7-9 cm, base flat, apex medium depressed, rind medium thin, tight, leathery, pitted, weakly pebbled, smooth, oil glands size medium, density medium, less conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Orange yellow, fine grained with an orange tinge, fibers medium, vesicles medium, juice high 42-45%, light golden, moderately acidic 0.9-1%, TSS 10-11%, fine rich flavor, aromatic.

Seed: Virtually seedless 0-4, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good cropper.

Leading Characters: Campbell Valencia: Introduction from California, late maturing sweet orange, fruit resemble Valencia almost in all respects, but the tree shows differences like more vigorous, more spiny than Valencia; good yielder but tends to alternate bearing, fruit size medium large, ellipsoid, orange yellow, rind medium thin, tight, pitted, flesh orange yellow, fine grained, high juicy, light golden, flavor good, a bit acid, aromatic.

CASAGRANDE

Type: Sweet Orange (*Citrus sinensis*); midseason, medium bearer, commercial variety grown on small scale.

Origin: Introduction from California 1973; Origin a seedling selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, moderately vigorous, upright, branches semi spreading, long spines (4-10 cm) on main stem and branches, water shoots thorny (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation weak to medium, length 7-9, width 3-4, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Orange-yellow, size medium, obovate, length 6-7, width 7-8 cm, base flat slightly depressed, apex flat, rind medium thin, medium tight, easy peel, surface rough, medium pitted, oil glands size medium, density medium, less prominent; albedo white, colomella medium, solid, segments 12-14.

Flesh and Flavor: Off-white, fibers medium, vesicles medium, juice high, 42-45%, color whitish creamy, acidity low 0.4-0.5%, TSS 9-10%, flavor rich, pleasant, very sweet, medium aromatic.

Seed: Few 5-10, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Casagrande: Introduction from California, medium bearer Sweet Orange, tree medium tall upright, moderately vigorous, long bold spines (4-10 cm) on main stem and branches, foliage green, medium dense; petiole wings absent, flowering once, solitary, fruit medium, obovate, orange yellow, rind medium thin, easy peel, surface rough, medium pitted (bigger in Valencia), oil glands medium, density medium, less conspicuous, flesh off-white, vesicles medium fibrous, juice high, whitish creamy, flavor rich, aromatic.

FISHER NAVEL

Type: Sweet Orange (*Citrus sinensis*); Fisher Navel, midseason medium bearer variety of navel orange.

Origin: Introduction from California 1973; **Origin.**

Maintainer: GPU **Sherkhana.**

Tree: Medium large, medium vigorous, branches semi spreading to spreading, water shoots almost spineless, (Rootstalk **Rough Lemon**).

Leaf: Foliage medium green, medium dense, leaf undulation weak, size medium, length 8-10, width 4-5, petiole 1-2 cm, petiole wings very small, narrow, aroma medium, refreshing.

Flower: Flowering season spring, anther creamy, aroma medium, pleasant.

Fruit: Deep orange, size medium to large, obovoid or elliptical, length 7-10, width 8-11 cm, base slightly collared, apex with medium to large nipple, rind medium thick, medium tight, surface rough, coarsely pitted, moderately pebbled, oil glands size medium, density medium, conspicuous; albedo white, columella medium, solid, segments 10-12.

Flesh and Flavor: Light yellow, firm, fine grained, fibers medium, vesicles tender, juice high 38-40%, color light creamy, sweet, low acidic 0.3-0.5%, TSS 9-10%, fine rich flavor, aromatic.

Seed: Virtually seedless 0-4, medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Fisher Navel: Introduction from California, a midseason, medium bearing variety of Navel orange, tree medium large, medium vigorous, shoots spineless, foliage medium green, medium dense, petiole wings very small, narrow, flowering once, fruit medium to large, obovoid to elliptical, color deep orange, rind medium thick, medium tight, coarsely pitted, moderately pebbled, navel present, oil glands size medium, density medium, medium conspicuous, fine grained, juice high, color light creamy, flavor excellent, sweet, aromatic.

FROST VALENCIA

Type: Sweet Orange (*Citrus sinensis*); Frost Valencia, Frost Nuceller Valencia, Nuceller Valencia, a late season high bearer variety of sweet orange.

Origin: Introduction from California 1973; Origin selection, a nuceller seedling released in 1952.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium large, vigorous, branches semi spreading, medium upright, water shoot spines rare, fruit stores well on tree, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, leaf undulation very weak, size large, length 8-12, width 4-6, petiole 1-2 cm, petiole wings absent, aroma refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Orange-yellow, size medium large, ellipsoid, length 7-9, width 8-10 cm, base flat, apex medium depressed, rind medium thick, medium tight, leathery, pitted, weakly pebbled, surface smooth, oil glands size medium, density medium, less conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Orange yellow, fine grained, fibers medium, vesicles medium, juice high 42-45%, color light golden, acidic 0.7-0.8%, TSS 9-11%, fine rich flavor, slightly acid, aromatic.

Seed: Virtually seedless 0-4, medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Frost Valencia: Introduction from California, late maturing, nuceller variety of sweet orange, good bearer, tree medium large, vigorous, shoot spines rare, foliage dark green, petiole wings absent, flowering once, fruit medium to large ellipsoid, color orange yellow, rind medium thick, medium tight, medium pitted, oil glands medium dense, less conspicuous, flesh orange yellow, fine grained, high juicy, light golden, virtually seedless, flavor good, aromatic.

JAFFA

Type: Sweet Orange (*Citrus sinensis*); Jaffa, Florida a midseason medium yielding, medium scale commercial variety of Sweet Orange.

Origin: Introduction from USA; Origin Palestine- a clonal selection of *Beledi seedling* group.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, medium vigorous, upright, branches semi spreading to upright, water shoots medium thorny, fruit hold on capacity poor, tends to alternate bearing, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, dense, leaf undulation very weak, length 8-10, width 4-6, petiole 1-2 cm, petiole wings absent, aroma refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Orange-yellow, size medium, globose to slightly elliptical or obovate, length 7-8, width 7-8 cm, base and apex round, rind medium thick, medium tight, peel well, surface finely pitted, medium pebbled, glossy, oil glands medium, density medium, prominent; albedo white, colomella medium, semi solid, segments 10-12, separating easily.

Flesh and Flavor: Light orange, fibers weak, vesicles medium, quick melting, juice high 40-45%, color whitish, acidity 0.5-0.55%, TSS 9-11%, aroma medium, pleasant, flavor rich, sweet.

Seed: Many, 8-12, medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Good bearer.

Leading Characters: Jaffa: Introduction from USA, originated in Palestine, medium bearer Sweet Orange, tree medium tall, medium vigorous, foliage dark green, dense, petiole wings absent, flowering once, flower solitary, fruit large globose or slightly elliptic, color orange yellow, rind medium thick, finely pitted, medium pebbled, medium tight, glossy, peel well, oil glands conspicuous, flesh light orange, segments separate easily, melting, juice high, color whitish, flavor fine, sweet, aromatic.

LANELATE NAVEL

Type: Sweet Orange (*Citrus sinensis*); Lanelate Navel, midseason medium bearer variety of navel orange.

Origin: Introduction from California 1973, **Origin.**

Maintainer: GPU, Sherkhana.

Tree: Medium large, medium vigorous, upright, branches semi spreading to upright, water shoots about spineless, (Rootstalk **Rough Lemon**).

Leaf: Foliage dark green, medium dense, leaf undulation weak, bigger, length 8-10, width 4-5, petiole 1-2 cm, petiole wings very small, aroma medium, refreshing.

Flower: Flowering season spring, anther creamy, aroma medium, pleasant.

Fruit: Orange yellow, size medium to large, obovoid or elliptical, length 7-9, width 8-10 cm, base sometimes slightly collared, apex with protruding medium to large nipple, rind medium thick, medium tight, surface rough, coarsely pitted, moderately pebbled, oil glands size medium, density medium, moderately conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Light yellowish, firm, fine grained, fibers medium, vesicles tender, juice high 38-40%, color light yellowish, sweet, low acidic 0.4-0.5%, TSS 9-10%, fine rich flavor, aromatic.

Seed: Seedless 0-2, medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Lanelate Navel: Introduction from California, a midseason, medium bearing variety of Navel orange, tree medium large, medium vigorous, upright, shoots spineless, foliage dark green, medium dense, petiole wings very small, flowering once, fruit medium to large, obovoid to elliptical, deep orange, rind medium thick, medium tight, coarsely pitted, moderately pebbled, oil glands medium conspicuous, fine grained, juice high, light creamy, flavor excellent, sweet, aromatic.

MARRS EARLY

Type: Sweet Orange (*Citrus sinensis*); medium early maturing, medium bearer heat tolerant, good variety.

Origin: Introduction from California 1973; Origin selection, a bud sport of Navel Orange trees.

Maintainer: ORI, Sargodha, GPU, Sherkhana.

Tree: Medium tall, bushy, medium vigorous, branches wide spreading, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, dense, leaf medium broad, leathery, undulation weak, length 7-9, width 3-5, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, fragrance medium, pleasant.

Fruit: Golden yellow, size medium, ovalish, length 10-11, width 11-12 cm, base round, apex flat, rind medium thin, medium tight, surface smooth, oil glands small, numerous, medium conspicuous; albedo white, colomella medium, solid, segments 10-14.

Flesh and Flavor: Creamy, fibers soft, vesicles medium melting, juice medium 35-38%, juice color creamy, acidity low 0.3-0.4%, TSS 10-12%, aroma medium, taste pleasant, sweet.

Seed: Virtually seedless 0-6, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Marrs Early: Introduction from California, a bud sport of Naval Orange tree, heat tolerant, medium bearer Sweet Orange, tree medium tall bushy, medium vigorous, medium spiny, foliage medium green, dense, leaf medium broad, leathery, petiole wings absent, flowering once, fruit size medium, round, color very attractive golden yellow, rind medium thin, medium tight, flesh color creamy, vesicles soft, medium melting, juice medium, color creamy, virtually seedless, flavor rich, taste very good, resemble Mosambi.

MORO

Type: Sweet Orange (*Citrus sinensis*); Moro, Moro Blood; very early maturing, medium bearer commercial variety grown on small scale.

Origin: Introduction from California 1973; an Italian variety, originated in Mediterranean Basin.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, bushy, resembles Mosambi, water shoots medium thorny, bearing generally in clusters of 2 or more, fruit hold on tree and storage well, (Rootstalk Rough Lemon).

Leaf: Foliage green, medium dense, leaf undulation very weak, length 7-9, width 4-5, petiole 1-2 cm, petiole wings very small, aroma refreshing.

Flower: Flowering season spring, solitary as well as bunches, anther yellow, fragrance medium, pleasant.

Fruit: Orange-yellow with light pink blush or red streaks, medium to large, round to obovoid, length 6-8, width 6-8 cm, base flat slightly depressed, apex flat, rind medium thick, medium tight, surface medium pebbled, oil glands size very small, density medium, medium prominent; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Deep pink almost violet red, fibers soft, vesicles tender, highly melting, juice medium 33-35%, pinkish, acidity low 0.4-0.6%, TSS 8-9%, aroma medium, flavor rich, sweet, distinctive, pleasant.

Seed: Few seeds 0-6, medium, white, polyembryonic, cotyledons whitish, inner coat light brown, chalazal spot purplish red.

Yield: Medium bearer.

Leading Characters: Moro: Italian variety of Mediterranean origin, introduced from California, tree medium tall upright, vigorous, foliage medium green, leaf wings narrow, fruit medium to large, round to obovoid, orange yellow, rind medium thick, flesh pinkish, juice medium, color pinkish, flavor rich, aromatic, pigments weak to medium on rind but medium to strong in the flesh, develops pigments early in high temperature.

MOSAMBI

Type: Sweet Orange (*Citrus sinensis*); Mosambi, Mozambique, early maturing, medium bearer, medium commercial popular variety.

Origin: Introduction from Mozambique; Origin unknown.

Maintainer: GPU, Sherkhana; ORI, Sargodha

Tree: Medium tall, vigorous, spreading wide, bushy, branches semi spreading but recurve under fruit load, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation very weak, length 8-10, width 4-5, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Light green at maturity, light-yellow at ripening, size medium, oblate, length 6-7, width 6-7 cm, base and apex rounded, clear circular ring on apex, rind medium thick, medium tight, peel not easy, surface rough, medium pebbled, slightly grooved longitudinally, oil glands size medium, density medium, prominent; albedo white, colomella medium, solid, segments 10-12, not separating easily.

Flesh and Flavor: pale yellow, fibers strong, vesicles strong, not melting, juice medium to high (38-43%), color whitish, acidity very low 0.2-0.35%, TSS 9-11%, flavor rich, taste very sweet, pleasant, aromatic.

Seed: Many seeded 5-30, medium, white, polyembryonic, cotyledons whitish, inner coat light brown, chalazal spot dark brown.

Yield: Medium to good bearer.

Leading Characters: **Mosambi**, Introduction from Mozambique (Africa) hence the name *Mosambi*, a popular medium bearer Sweet Orange, tree medium tall, vigorous, foliage dark green, flowering once, fruit size medium oblate, light green to orange yellow on ripening, rind medium thick, medium tight, flesh pale yellow, seeds many, vesicles strong fibrous non melting, juice medium to high, whitish, flavor good, very sweet, mild aromatic.

OLINDA VALENCIA

Type: Sweet Orange (*Citrus sinensis*); Olinda Valencia, a late season high bearer variety grown on small scale.

Origin: Introduction from California 1973; Origin probably a chance seedling selection of Valencia.

Maintainer: ORI, Sargodha.

Tree: Medium large, vigorous, branches semi spreading to spreading, water shoots medium thorny, tends to be alternate bearing, fruit hold on tree very well, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation very weak, size large, length 8-11, width 4-6, petiole 1-2 cm, petiole wings narrow, aroma medium, refreshing, flush light green.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Orange-yellow sometimes with greenish tinge, medium large, ellipsoid, length 7-9, width 7-9 cm, base flat, apex medium depressed, rind medium thick, tight, leathery, finely pitted, weakly pebbled, smooth, oil glands size medium, density medium, less conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Orange yellow, fine grained with an orange tinge, fibers medium, vesicles medium, juice high 44-47%, light golden, acidic 0.8-1.0%, TSS 10-11%, fine rich flavor and aromatic, taste slightly acidic.

Seed: Few seeded 0-5, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Olinda Valencia: Introduction from California, late maturing sweet orange, good bearer, tends to alternate bearing, tree medium large, shoots medium thorny, foliage medium green, petiole wings narrow, fruit medium large ellipsoid, color orange yellow, rind medium thick, tight, finely pitted, flesh orange yellow, fine grained, juice high, color light golden, a bit acidic, flavor rich, aromatic.

PARSON

Type: Sweet Orange (*Citrus sinensis*); Parson, Parson Brown, an early season medium bearer variety of sweet orange.

Origin: Introduction from Florida 1973; Origin a chance seedling selection in Florida.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium large, vigorous, medium upright, branches semi spreading, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, size large, length 8-11, width 4-5, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Orange-yellow, size medium large, globose, length 7-9, width 8-10 cm, base with short radial furrow, apex with slight circular ring, medium depressed, rind medium thick, medium tight, surface finely pitted, moderately pebbled, oil glands size medium, density medium, medium conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Dull orange, firm, fine grained, fibers medium, vesicles melting, juice high 38-40%, light creamy, sweet, low acidic 0.7-0.8%, TSS 9-10%, fine rich flavor, aromatic.

Seed: Fairly seedy 10-20, medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Moderate cropper.

Leading Characters: Parsen: Introduction from Florida, an early maturing variety of sweet orange, medium bearer, tree medium large, vigorous, shoots medium thorny, foliage dark green, medium dense, petiole wings absent, flowering once, fruit medium large, globose, orange yellow, rind medium thick, finely pitted, moderately pebbled, oil glands medium, medium conspicuous, flesh dull orange, fine grained, juice high, light creamy, flavor good, aromatic.

PINEAPPLE

Type: Sweet Orange (*Citrus sinensis*); Pineapple, midseason, good yielding, medium scale variety.

Origin: Introduction from California 1973; Origin a chance seedling selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, medium vigorous, spreading upright, branches upright to semi spreading, water shoot thorns absent, hold on capacity poor to medium, tends to alternate bearing, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation very weak, length 8-10, width 4-6, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Golden Orange, size medium, spherical to slightly obovate, length 6-8, width 7-9 cm, base and apex round, apex with weak ring, rind medium thick, peel well, surface finely pitted, slightly pebbled, glossy, oil glands medium, density medium, prominent; albedo white, colomella medium, semi solid, segments 10-12, separating easily.

Flesh and Flavor: Golden yellow, fibers weak, vesicles medium tender, quick melting, juice high 40-45%, color whitish, acidity 0.45-0.55%, TSS 10-11%, aroma medium, pleasant, flavor rich, sweet.

Seed: Medium seedy 6-10, medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Pineapple: Introduction from California, a chance seedling selection, medium bearer very attractive Sweet Orange, tree medium tall, medium vigorous, foliage dark green, medium dense, petiole wings absent, fruit medium, spherical to slightly obovate, golden orange, rind medium thick, slightly pebbled, peel easy, oil glands medium dense, conspicuous, flesh golden yellow, segments separating easily, juice high, whitish, flavor rich, sweet, aromatic.

RUBY BLOOD

Type: Sweet Orange (*Citrus sinensis*); Ruby Blood, Ruby; mid to late season, medium bearer commercial variety grown on large scale.

Origin: Introduction from California 1973; Origin Mediterranean.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium tall, vigorous, compact, branches semi spreading but recurve under fruit load, water shoots medium spiny, spines very small, fruit hold on and storage very good (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, size large, length 10-13, width 5-7, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Deep orange, red flushed on sides under favorable conditions, size medium, globose to slightly oblong, length 6-7, width 7-8 cm, base flat, apex with circular furrow or ridge, medium depressed, rind medium thick, finely pitted, lightly pebbled, oil glands size medium, density medium, moderately conspicuous; albedo white, colomella size medium, solid, segments 10-12.

Flesh and Flavor: Orange occasionally with red streaks, tender, vesicles medium, fine grained, juice high 40-45%, light reddish, acidity 0.5-0.6%, TSS 9-11%, flavor rich, fine, aromatic.

Seed: Few 4-10, medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Good bearer.

Leading Characters: Ruby Blood: Introduction from California, mid to late sweet orange, tree medium tall, vigorous, compact, foliage dark green, size large, fruit medium, globose, deep orange with red flushed on sides, rind medium thick, finely pitted, flesh orange occasionally with red streaks, fine grained, juice high, light reddish, flavor rich, fine, refreshing. Development of color is highly variable and uncertain under variable conditions.

SALUSTIANA

Type: Sweet Orange (*Citrus sinensis*); early maturing, medium bearer variety grown on medium scale Sweet Orange.

Origin: Introduction from Spain 1988; Origin a limb sport selection in Spain.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, spreading medium upright, open, branches semi spreading, water shoots medium thorny, fruit hold on tree and storage good (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation absent to weak, length 7-9, width 4-5, petiole 1-2 cm, petiole wings very short and narrow, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Orange-yellow, size medium large, roundish to sub-globose, length 8-10, width 7-9 cm, base flat with medium depression and weak radial furrows, apex flat, medium depressed with weak to medium circular ring, rind medium thick, medium loose, peel medium adherent, surface medium pebbled, oil glands medium, density medium, less prominent; albedo white, columella medium, hollow, segments 10-12.

Flesh and Flavor: Off-white, fibers medium, vesicles medium soft, melting, juice high 40-45%, color whitish, acidity low 0.35-0.45%, TSS 10-12%, aroma medium, flavor rich, very pleasant.

Seed: Virtually Seedless, 0-6, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Salustiana: Introduction from Spain, medium bearer Sweet Orange, tree medium tall, upright, vigorous, foliage dark green, medium dense; leaf wings short and narrow, flowering once, flower single, fruit medium, roundish to sub-globose, orange yellow, rind medium thick, peel medium adherent, albedo white, flesh off-white, melting, juice high, whitish, flavor rich, sweet, strong aromatic.

SANGUINELLI

Type: Sweet Orange (*Citrus sinensis*); Sanguinelli, Spanish Sanguinelli, Sanguinella; early season medium bearer grown on medium scale.

Origin: Introduction from Italy; Origin selection a limb sport of Doblefina.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, medium vigorous, low tolerance of heat, upright, medium spreading, water shoots almost thornless, fruit mostly borne inside in medium to large clusters and solitary on branches, fruit hold on tree and store good, (Rootstalk Rough Lemon).

Leaf: Foliage light green, medium dense, leaf undulation very weak, length 7-8, width 4-5, petiole 1-2 cm, mid rib prominent, petiole wings very small, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Reddish green with apical portion blushed deep red, size medium to large, very attractive on branches, round to oblong, length 6-7, width 7-8 cm, base flat, medium depressed, apex flat, rind medium thick, medium loose, medium pebbled, oil glands medium, sparse, medium conspicuous; albedo pinkish, colomella solid, segments 10-12.

Flesh and Flavor: Uniform red, very attractive color streaks, fibers medium strong, vesicles large, tight, juice medium 35-38%, color reddish, acidity 0.45-0.55%, TSS 9-10%, flavor pleasant, aromatic.

Seed: virtually seed less 0-6, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown, chalazal spot purplish red.

Yield: Medium bearer.

Leading Characters: Sanguinelli: Introduction from Italy, medium cropper, tree medium large, upright, thornless, foliage light green, petiole wings very small, fruit medium to large, round to oblong, pigmented, blushed deep red, fruit mostly borne inside in bunches on main stem and solitary on branches, rind thick, medium loose, flesh red, juice high, reddish pigmented, flavor fine, rich, tasty, aromatic.

SPRING NAVEL

Type: Sweet Orange (*Citrus sinensis*); Spring Navel, midseason medium bearer variety of navel orange.

Origin: Introduction from California 1973,

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium short, medium vigorous, branches semi spreading, medium upright, water shoots almost spineless, (Rootstalk Flying Dragon).

Leaf: Foliage medium green, medium dense, leaf undulation weak, leaf bigger, length 10-12, width 5-7, petiole 1-2 cm, petiole wings narrow, aroma mild, refreshing.

Flower: Flowering season spring, anther creamy, aroma medium, pleasant.

Fruit: Deep orange, size medium to large, spherical to obovoid or elliptical, length 7-10, width 8-11 cm, base sometimes slightly collared, apex with protruding medium to large and broad nipple, rind medium thick, medium tight, rough, coarsely pitted, moderately pebbled, oil glands size medium, density medium, medium conspicuous; albedo white, columella medium, solid, segments 10-12.

Flesh and Flavor: Creamy yellow, firm, fine grained, fibers medium, vesicles melting, juice high 38-40%, color light yellow, sweet, low acidic 0.4-0.5%, TSS 9-10%, flavor fine, rich, aromatic.

Seed: Seedless 0-2, medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Spring Navel: Introduction from California, a midseason, medium bearing variety of Navel Orange, tree medium short, medium vigorous, shoots spineless, foliage medium green, medium dense, petiole wings narrow, fruit medium to large, spherical, obovoid to elliptical, color deep orange, rind medium thick, medium tight, coarsely pitted, moderately pebbled, oil glands medium conspicuous, juice high, color light creamy, flavor excellent, fine, taste sweet, aromatic, almost seedless.

SUCCARI

Type: Sweet Orange Acidless Sweet Orange, (*Citrus sinensis*); Succari, Sukkary, *Shakri Malta*; early season, medium bearer acidless Sweet Orange grown on small scale.

Origin: Introduction; Origin unknown, a popular variety of Egypt.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, moderately vigorous, branches semi spreading, water shoots spineless (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation weak, length 6-8, width 3-5, petiole 1-2 cm, petiole wings narrow to medium, aroma medium, refreshing.

Flower: Flowering spring, anther yellow, fragrance medium, pleasant.

Fruit: Orange-yellow, size medium, obovate to globose, length 7-9, width 6-8 cm, base flat slightly depressed with slight ridges, apex flat, rind medium thick, medium tight, peel medium, surface medium smooth, oil glands size small, density medium, less prominent; albedo white, colomella medium, solid, segments 10-12, separate easily.

Flesh and Flavor: Off-white, pale, fibers medium, vesicles strong, not melting, juice medium 35-38%, color whitish, acidity very low 0.18-0.2%, TSS 9-10%, aroma medium, taste very sweet.

Seed: Many seeded, 20-25, size bold, white, polyembryonic, cotyledons whitish, inner coat light brown, chalazal spot creamy.

Yield: Medium bearer.

Leading Characters: Succari or Shakri Malta: Introduction, a popular acidless variety of Egypt, early season, medium bearer, tree medium tall, moderately vigorous, foliage medium green, density sparse as compared to Mosambi, petiole wings narrow to medium, flowering once, fruit medium, obovate to globose, color orange yellow, rind medium thick, medium smooth, medium tight, oil glands less prominent, flesh off-white, vesicles not melting, juice medium, whitish, flavor good, very sweet and aromatic.

SUGAR MALTA

Type: Sweet Orange, Acidless Sweet Orange, (*Citrus sinensis*); *Sugar malta* early season, medium bearer variety grown on small scale.

Origin: Introduction; Origin unknown.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, moderately vigorous, spreading upright, branches semi spreading, water shoots spineless (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation weak, leaf length 7-9, width 3-5, petiole 1-2 cm, petiole wings narrow, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, fragrance medium, pleasant.

Fruit: Orange-yellow, size medium, obovate to oblongish, length 7-9, width 7-8 cm, base flat slightly depressed with slight ridges, apex flat, rind thick, medium to tight, peel medium tight, surface medium smooth, oil glands size small, density medium, less prominent; albedo white, colomella medium, solid, segments 10-12, separate easily.

Flesh and Flavor: Whitish, fibers medium, vesicles strong, not melting, juice medium 35-40%, color whitish, acidity very low 0.1-0.2%, TSS 10-11%, aroma medium, flavor good, very sweet.

Seed: Many seeded 20-25, size medium bold, white, polyembryonic, cotyledons whitish, inner coat light brown, chalazal spot creamy.

Yield: Medium bearer.

Leading Characters: Sugar Malta: Introduction, origin unknown, early season, medium bearer acidless orange, tree medium tall, moderately vigorous, foliage medium green, medium dense, petiole wings narrow, flowering once, fruit length more than Succari, shape obovate to oblongish, color orange yellow, rind thicker than Succari, medium smooth, peel medium tight, oil glands size small, less prominent, flesh whitish, many seeded, vesicles not melting, juice medium, color whitish, flavor excellent, very sweet, medium aromatic.

TAROCCO

Type: Sweet Orange (*Citrus sinensis*); midseason, medium bearer commercial variety grown on small scale.

Origin: Introduction from Italy 1988; Origin clonal selection in Italy.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Tall growing, vigorous, spreading upright, branches long, water shoots medium thorny, fruit hold on capacity poor but storage well, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium lax, leaf undulation weak, sharp pointed, length 7-10, width 4-5, petiole 1-2 cm, petiole wings narrow, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, fragrance medium, pleasant.

Fruit: Orange-yellow blushed with red on ripening, size medium, ob-ovate to globose, length 6-8, width 7-8 cm, base round, furrowed basal collar, apex flat slightly depressed, rind thin, medium loose, surface medium pebbled, depressed, glossy, oil glands size medium, density medium, medium prominent; albedo white, colomella medium, solid, segments 10-12, large and free.

Flesh and Flavor: Orange yellow red pigmented, fibers medium firm, vesicles medium, juice high 42-44%, color whitish, acidity low 0.45-0.55%, TSS 9-10%, flavor rich, tasty, aromatic.

Seed: Virtually seedless 0-6, medium, white, polyembryonic, cotyledons light green, inner coat light brown, chalazal spot purplish red.

Yield: Medium bearer.

Leading Characters: Tarocco: Introduction from Italy, medium bearer pigmented sweet orange, tree tall growing, spreading upright, long irregular branches, foliage dark green, medium lax; petiole wings narrow, flowering once, fruit medium, obovate to globose, orange yellow with red blush, rind thin, medium loose, pebbled, flesh orange yellow, red pigmented, firm, juice high, flavor rich, tasty, aromatic.

VALENCIA LATE

Type: Sweet Orange (*Citrus sinensis*); Valencia, Valencia late, a very late season high bearer Sweet Orange grown on sizeable scale.

Origin: Introduction from California 1973; Origin probably Portugal as a natural selection.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium tall, vigorous, branches semi spreading upright, water shoots medium thorny, tends to alternate bearing especially if fruit are harvested very late in the season; fruit hold on tree well (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, length 8-11, width 4-6, petiole 1-2 cm, petiole wings small, aroma refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Orange-yellow sometimes with greenish tinge, size medium large, ellipsoid, length 7-9, width 7-9 cm, base flat, apex flat, medium depressed, rind medium thin, tight, leathery, pitted, weakly pebbled, smooth, oil glands size medium, less conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Orange yellow, fine grained with an orange tinge, fibers medium, vesicles medium, juice high 42-45%, color light golden, moderately acidic 0.9-0.95%, TSS 10-11%, fine rich flavor and aromatic, taste slightly acidic, excellent for processing.

Seed: Virtually seedless 0-4, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Valencia Late: Introduction from California, late to very late maturing sweet orange, fruit can stay on the tree for longer periods, tree medium tall, foliage dark green, leaf petiole wings small, flowering once, fruit large ellipsoid, orange yellow, rind medium thin, pitted (pits bigger than Casagrande) flesh orange yellow, fine grained, high juice high, color light golden, flavor good, a bit acid, aromatic.

WASHINGTON NAVEL

Type: Sweet Orange, Navel Orange (*Citrus sinensis*); Washington Navel, midseason, medium bearing variety of navel orange.

Origin: Introduction from California 1973; Origin unknown.

Maintainer: ORI, Sargodha; GPU, Sherkhana.

Tree: Medium large, medium vigorous, branches medium droopy, medium upright, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, size large, length 8-12, width 4-5, petiole 1-2 cm, petiole wings narrow, aroma medium, refreshing.

Flower: Flowering season spring, anther creamy, aroma medium, pleasant.

Fruit: Deep orange, medium to large, spherical to obovoid or elliptical, length 8-11, width 9-12 cm, base sometimes slightly collared, apex with protruding medium to large and broad nipple, rind medium thick, medium tight, surface rough, coarsely pitted, moderately pebbled, oil glands size medium, density medium, medium conspicuous; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Light creamy, firm, fine grained, fibers medium, vesicles melting, juice high 38-40%, color light creamy, sweet, low acidic 0.4-0.5%, TSS 9-10%, fine rich flavor, aromatic.

Seed: Seedless 0-2, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Washington Navel: Introduction from California, a midseason, medium bearing variety of Navel orange, tree medium large, medium vigorous, shoots medium thorny, foliage dark green, medium dense, petiole wings narrow, fruit medium to large, spherical, obovoid to elliptical, deep orange, rind medium thick, medium tight, coarsely pitted, moderately pebbled, navel present, oil glands medium conspicuous, juice high, light creamy, flavor excellent, aromatic.

Citrus varieties

Grape Fruits

GRAPE FRUITS

BUNTAN

Type: Pummelo (*Citrus grandees*); Buntan Pummelo, *Chakotra*, a mid late season, medium bearer variety grown on small scale.

Origin: Introduction from California 1973; Origin a chance seedling selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Large growing, moderately vigorous, spreading wide, branches droopy, water shoots thorny, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, dense, leaf undulation weak, length 8-11, width 4-6, petiole 1-2 cm, petiole wings large, aroma mild.

Flower: Flowering spring, mostly solitary, anther yellow, aroma strong.

Fruit: Yellow, size very large, ovate depressed on both ends, base more depressed than apex, length 10-15, width 13-19 cm, rind very thick, medium tight, very rough pebbled, oil glands bigger, medium, conspicuous; albedo pinkish, colomella hollow, segments 14-16.

Flesh and Flavor: Light pink, fibers hard, vesicles large, hard, juice low 32-35% color light pink, acidity low 0.18-0.20%, TSS 8-9%, taste good, traces of bitter, flavor fair.

Seed: Many seeded 40-50, size large to very large, white, monoembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Buntan Pummelo: Introduction from USA, medium bearer variety of Pummelo (*Chakotra*), tree wide spreading, branches spreading, droopy, foliage dark green, dense, leaves large, petiole wings large, flowering season spring, fruit size very large, ovate depressed on both ends, color yellow, rind very thick, medium tight, surface very rough, strongly pebbled, oil glands bold, density medium, conspicuous, albedo color pinkish, flesh color light pink, vesicle size large, fibers hard, juice low, color light pink, taste somewhat bitter, flavor fair, seeds many, monoembryonic as compared to Grape Fruits and Sweet Oranges which are polyembryonic.

DUNCAN

Type: Grapefruit (*Citrus paradisi*), Duncan, a medium early, good bearer commercial variety of Grape Fruit.

Origin: Introduction from USA; a chance seedling selection (1892).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, spreading wide, branches long, medium droopy, semi spreading, water shoots medium thorny, fruit hold on tree and storage good, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation very weak, medium, length 7-9, width 4-5, petiole 1-1.5 cm, petiole wings medium broad, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow, size medium, oblate to globose, length 9-11, width 10-12 cm, base round, with short furrow, depressed, apex flat, areola faint, rind medium thick, leathery, medium tight, surface finely pitted, smooth, oil glands medium, density medium, inconspicuous; albedo whitish, colomella medium, semi solid, segments 10-14.

Flesh and Flavor: Light yellow, fibers medium, vesicles medium large, medium free, juice medium 35-39%, color light yellow, acidity high 1.0-1.5%, TSS 9-10%, flavor fine rich sweetish bitter, a little acid, aroma medium, pleasant.

Seed: Few seeded 4-10, size medium to large, dusky white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Duncan: Introduction from USA, medium early grape fruit, tree medium tall, branches spreading, medium droopy, foliage dark green, medium dense, petiole wings medium broad, flowering once, fruit medium, oblate to globose, rind medium thick, leathery, oil glands medium bold, density medium, inconspicuous, flesh light yellow, medium juicy, color light yellow, flavor rich, sweetish bitter aromatic.

FLAME

Type: Grapefruit (*Citrus paradisi*); Flame, an early season medium bearer variety grown on medium scale.

Origin: Introduction from USA; Origin a limb sport.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, bushy, branches long drooping, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation weak, length 7-9, width 4-5, petiole 1-2 cm, mid rib prominent, petiole wings large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with beautiful red blush, size medium, roundish, length 7-9, width 8-10 cm, base round, apex flat slightly depressed with weak circular ring, rind thick, medium loose, medium smooth, oil glands size medium, density medium, conspicuous; albedo pinkish, colomella solid, segments 10-14.

Flesh and Flavor: Flame red, fibers medium strong, vesicles large, tight, juice high 40-42%, juice color beautiful pinkish, acidity high 1.0-1.3%, TSS 7-8%, taste good, flavor pleasant.

Seed: virtually seed less 0-5, size large, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Flame: Introduction from USA, early season, medium bearer Grapefruit, tree medium tall, vigorous, branches long droopy, foliage medium green, medium dense, petiole wings large, flowering once, fruit size medium, roundish, color light yellow with reddish blush (resembling flame), rind thick, medium loose, medium smooth, apex with weak circular ring, oil glands size medium, density medium, conspicuous, virtually seedless, flesh color flame red, juice high, color beautiful pinkish, flavor, fine rich pleasantly aromatic.

FROST MARSH

Type: Grapefruit (*Citrus paradisi*); Frost Marsh, an early season good cropper variety grown on medium scale.

Origin: Introduction from California 1973; Origin selection, a nuceller seedling of Marsh Seedless.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall growing, vigorous, branches spreading, water shoots long thorny, fruit hold on tree very good but with some fading of fruit and flesh color, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, dense, leaf undulation weak, length 8-10, width 5-6, petiole 1-2 cm, petiole wings large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow, size medium large, globose to spherical, length 9-11, width 8-10 cm, base flat, apex slightly depressed, rind medium thick, medium loose, surface smooth, oil glands size large, dense, conspicuous; albedo pinkish, colomella hollow, segments 12-14.

Flesh and Flavor: Whitish creamy, fibers medium strong, vesicles large, tight, juice high 46-48%, color off-white, flavor pleasant, acidity 1.0-1.3%, TSS 8-9%, flavor fine, a bit acid, pleasant, aromatic.

Seed: Virtually seedless, 0-6, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Frost Marsh: Introduction from California; a nuceller seedling of Marsh Seedless, good bearer, early season Grapefruit, tree medium tall growing, vigorous, branching spreading, foliage dark green, dense, petiole wings large, flowering spring, fruit size medium large, globose to spherical, color light yellow, base flat, apex slightly depressed, rind medium thick, medium loose, surface smooth, oil glands size large, dense, moderately conspicuous, albedo pinkish, flesh whitish creamy, vesicles large, juice high, color off-white, flavor fine, rich, aromatic.

MARSH JB 430

Type: Grapefruit (*Citrus paradisi*); Marsh JB 430, a midseason, good bearer variety grown on medium scale.

Origin: Introduction from California 1973; Origin selection

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, branches spreading, water shoots medium thorny, fruit bearing occasionally bunches from main, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation weak, size medium to large length 7-11, width 4-6, petiole 1-2 cm, petiole wings large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with red blush, size medium, obovoid to globose, length 8-10, width 10-12 cm, base flat, apex flat, slightly depressed, slight ridges on base, rind medium thick, medium loose, smooth, oil glands size small, density numerous, less conspicuous; albedo pinkish, colomella hollow, segments 12-14.

Flesh and Flavor: Whitish with red streaks, fibers medium strong, vesicles large, tight, juice high 40-43%, color reddish, acidity high 1.5-1.8%, TSS 7-8%, taste good, flavor rich, pleasant, aromatic.

Seed: Seeds few 1-8, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Marsh JB 430: Introduction from California, a midseason, medium bearer Grapefruit, tree medium tall, vigorous, spreading, foliage medium green, medium dense, petiole wings large, flowering spring, fruit bearing solitary, occasionally bunches from main stem, size medium, obovoid to globose, color light yellow, rind medium thick, medium loose, smooth, oil glands small, density numerous, less conspicuous, albedo pinkish, flesh whitish with red streaks, vesicles large, tight, juice high, reddish, flavor rich, aromatic.

MARSH SEEDLESS

Type: Grapefruit (*Citrus paradisi*), Marsh, Marsh Seedless; a late season medium bearer variety grown on small scale.

Origin: Introduction from Florida 1973; Origin a chance seedling selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, spreading wide, branches semi spreading but recurve under fruit load, water shoots medium thorny, fruit hold on tree and storage very good, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation very weak, size medium, length 7-9, width 4-5, petiole 1-1.5 cm, petiole wings short and broad, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow, size medium, oblate to spherical, length 9-11, width 10-12 cm, base round, narrow, depressed, apex flat, areola distinct, rind medium thick, leathery, medium tight, surface finely pitted, smooth, oil glands medium, density medium, conspicuous; albedo yellowish, colomella medium, semi solid, segments 10-12.

Flesh and Flavor: Color buff, fibers medium, vesicles large, tender, medium free, juice high 38-40%, color light yellow, acidity 1.4-1.6%, TSS 9-11%, flavor fine rich sweet, a bit acid, pleasant, aromatic.

Seed: Virtually seedless, 0-6, size medium to large, dusky white, polyembryonic, cotyledons light blue, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Marsh Seedless: Introduction from Florida, a chance seedling selection, high yielder grape fruit, tree medium tall growing, foliage dark green, medium dense, petiole wings short, broad, flowering spring, fruit size medium, oblate to spherical, rind medium thick, leathery smooth, medium tight, oil glands size medium bold, density medium, conspicuous, flesh buff yellow, juice high, light yellow, flavor fine, rich, a bit acidic, aromatic.

PINK RUBY

Type: Grapefruit (*Citrus paradisi*); Pink Ruby, a midseason, medium bearer variety of attractive colorful fruits.

Origin: Introduction from France; Origin selection.

Maintainer: GPU, Sherkhana.

Tree: Medium tall, bushy, vigorous, branches long drooping, water shoots less spiny, (Rootstalk **Rough Lemon**).

Leaf: Foliage dark green, dense, leaf undulation weak, concave, length 9-11, width 4-5, petiole 1-2 cm, mid rib prominent, petiole wings medium to large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with pink blush, size medium to large, roundish to obovoid, length 7-10, width 8-11 cm, base round, apex flat slightly depressed with prominent circular furrow, rind medium thick, medium loose, medium smooth, oil glands size medium, density medium, conspicuous; albedo pinkish, colomella solid, segments 10-14.

Flesh and Flavor: Reddish pink, fibers medium strong, vesicles large, tight, juice high 35-40%, juice color pinkish, acidity high 1.0-1.3%, TSS 7-8%, taste good, flavor pleasant.

Seed: virtually seed less 0-5, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Pink Ruby: Introduction from France, a midseason, medium bearer, colorful Grapefruit, tree medium large, bushy spreading, foliage dark green, dense, petiole wings medium to large, flowering spring, fruit size medium to large, roundish to obovoid, color light yellow with pink blush, rind thick, medium loose, medium smooth, apex with prominent circular furrow, oil glands size medium, density dense, conspicuous, virtually seedless, flesh reddish pink, juice high, color beautiful pinkish, flavor rich, pleasantly aromatic.

RAY RUBY

Type: Grapefruit (*Citrus paradisi*); Ray Ruby, a midseason season, medium cropper variety of attractive colorful fruits.

Origin: Introduction from France; Origin selection.

Maintainer: GPU, Sherkhana.

Tree: Medium height, bushy, vigorous, branches medium spreading, water shoots medium thorny (Rootstalk **Rough Lemon**).

Leaf: Foliage dark green, medium dense, leaf undulation weak, length 8-10, width 4-6, petiole 1-2 cm, petiole wings medium large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with red blush, size medium large, roundish to spherical, length 7-9, width 8-10 cm, base round, apex flat slightly depressed with weak circular furrow, rind thick, medium loose, medium smooth, oil glands size medium, density medium, conspicuous; albedo pinkish, colomella solid, segments 10-14.

Flesh and Flavor: Reddish, fibers medium strong, vesicles large, tight, juice high 35-40%, juice color beautiful pinkish, acidity high 1.0-1.5%, TSS 7-9%, taste good, flavor pleasant.

Seed: virtually seed less 0-5, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Ray Ruby: Introduction from France, a midseason, medium bearer variety of Grapefruit, tree medium large, bushy, medium spreading, foliage dark green, medium dense, petiole wings medium large, flowering spring, fruit size medium large, shape roundish to spherical, color light yellow with reddish blush, rind thick, medium loose, surface medium smooth, apex flat with weak circular ring, oil glands size medium, density medium, conspicuous, flesh reddish, juice high, color pinkish, flavor rich pleasantly aromatic.

REDBLUSH

Type: Grapefruit (*Citrus paradisi*); Redblush, an early to midseason, medium yielding, small scale variety.

Origin: Introduction from California 1973; Origin selection

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, branches semi spreading but recurve under fruit load, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium dense, leaf undulation weak, broad, length 6-8, width 4-6, petiole 1-2 cm, petiole wings medium large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: yellow with red blush, size medium to large, round to oval, length 9-11, width 10-12 cm, base round, apex flat, depressed, rind thick, medium loose, surface finely pitted, medium smooth, oil glands size medium, density medium, medium conspicuous; albedo pinkish, colomella medium, semi solid, segments 10-14.

Flesh and Flavor: Red, fibers strong, vesicles large, tight, juice medium 35-38%, acidity high 1.0-1.4%, TSS 9-10%, juice color beautiful red, taste good, flavor pleasant.

Seed: Seedless 0-4, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Redblush: Introduction from California, early to midseason, medium bearer beautiful Grapefruit, tree medium tall, branches semi spreading, foliage medium green, medium dense, petiole wings medium large, flowering spring, fruit size medium to large, round to oval, color yellow with attractive red blush (resemble Shamber but skin red blushed), rind thick, medium loose, albedo pinkish, oil glands size medium, density medium, moderately conspicuous, flesh red, medium juicy, striking red, flavor fine, rich, tasty, aromatic.

REED

Type: Grapefruit (*Citrus paradisi*); Reed, an early to midseason, medium bearer variety grown on medium scale.

Origin: Introduction from California 1973; Origin selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous, branches spreading but recurve under fruit load, water shoots less thorny, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, length 7-9, width 4-5, petiole 1-2 cm, petiole wings medium to large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with red blush, size medium to large, obovoid to globose, length 9-11, width 10-12 cm, base flat, apex slightly depressed, rind thick, medium loose, surface smooth, oil glands size large, dense, conspicuous; albedo pinkish, colomella hollow, segments 12-14.

Flesh and Flavor: Whitish creamy, fibers medium strong, vesicles large, tight, juice high 40-42%, color creamy whitish, acidity high 1.0-1.2%, TSS 8-10%, taste good, flavor pleasant, aromatic.

Seed: Seeds few, 1-8, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Reed: Introduction from California, an early to midseason, medium bearer grapefruit, tree medium large, vigorous, branches spreading, foliage dark green, medium dense, petiole wings medium to large, flowering spring, fruit size medium to large, obovoid to globose, color light yellow with red blush, apex slightly depressed, rind thick, medium loose, surface smooth, oil glands size large, dense, conspicuous, albedo pinkish, flesh whitish creamy, fibers medium strong, vesicles large, juice high, color creamy whitish, acidity high, flavor fine, rich, taste good, aromatic.

RUBY

Type: Grapefruit (*Citrus paradisi*); Ruby, Red Seedless, a midseason, medium bearer variety grown on medium scale.

Origin: Introduction from California 1973; Originated as limb sport of Thompson, probably the first citrus variety planted in USA.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, vigorous bushy, branches spreading, water shoots almost spineless, , fruit hold on tree well but with some fading of flesh color. (Rootstalk Rough Lemon).

Leaf: Foliage dark green, glossy, medium dense, leaf undulation absent to weak, length 7-9, width 3-4, petiole 1-2 cm, petiole wings small but broad, aroma medium, refreshing.

Flower: Flowering spring, single, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with red crimson blush, medium to large, oblate to spherical, length 9-11, width 8-10 cm, base round little tapering, short necked, apex flat, rind medium thin, fairly loose, easy peel, medium smooth, slightly ridged, oil glands bold, density sparse but dense towards base, conspicuous, albedo pigmented pinkish, colomella medium hollow, segments 10-12.

Flesh and Flavor: Pinkish, fibers medium hard, vesicles large, tight, juice high 47-50%, color pinkish, acidity high 1.3-1.4%, TSS 9-10%, flavor, rich, aroma pleasant.

Seed: virtually seedless, 0-5, medium, dusky white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Ruby: Introduction from California, medium yielder Grapefruit, tree medium tall, spreading, foliage dark green, glossy, fruit medium to large , oblate to spherical, light yellow with red crimson blush, rind medium thin, fairly loose, slightly ridged, base short necked, oil glands bold, flesh pinkish, fiber medium hard, juice high, flavor fine, rich, aromatic.

SHAMBER

Type: Grapefruit (*Citrus paradisi*); Shamber, an early to midseason medium bearer variety grown on medium scale.

Origin: Introduction from California 1973, originated as limb sport of Marsh Seedless.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium large, medium vigorous, branches spreading but recurve under fruit load, water shoots medium thorny, fruit hold on tree well, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, glossy, medium to dense, leaf undulation weak, length 8-10, width 5-6, petiole 1-2 cm, petiole wings medium large, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with red crimson blush, size medium to large, globose, length 8-9, width 9-10 cm, base and apex flat slightly depressed, slight ridges on base, rind medium thick, medium loose, surface very smooth, oil glands size bold, medium dense, conspicuous; albedo pinkish, colomella hollow, segments 12-14.

Flesh and Flavor: Red (color less than Star Flame), fibers medium strong, vesicles large, tight, juice high 48-49.5%, color reddish, acidity high 1.3-1.5%, TSS 9-10%, flavor rich, pleasant, aromatic.

Seed: Seedless 0-6, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Shamber: Introduction from California, early to midseason, medium bearer Grapefruit, tree medium large spreading, foliage dark green, glossy, leaf wings medium large, fruit medium to large, globose, light yellow with red crimson blush, rind medium thick, medium loose, fairly smooth, oil glands large, dense, conspicuous, flesh red, juice high, color reddish, flavor rich, aroma pleasant, taste good.

STAR RUBY

Type: Grapefruit (*Citrus paradisi*); Star Ruby, an early to midseason medium bearer variety grown on medium scale.

Origin: Introduction from California 1973; Origin selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium dwarf, bushy, medium vigorous, branches drooping, chlorotic leaves due to sun burning, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, very dense, leaf undulation absent to weak, length 7-9, width 3-4, petiole 1-2 cm, petiole wings large, show blotchy chlorotic areas most probably due to heat, aroma medium, refreshing.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow with dark pink blush, rather deep red, size medium to large, round to globose, length 8-10, width 9-11 cm, base round, apex flat, rind medium thin, medium loose, medium smooth, oil glands bold, dense towards base, conspicuous; albedo pinkish, colomella medium solid, segments 10-14.

Flesh and Flavor: Deep red pinky, fibers medium strong, vesicles large, tight, juice high 40-42%, color dark reddish, acidity high 1.5-1.8%, TSS 9-11%, , flavor rich, aroma pleasant.

Seed: Virtually seedless 0-6, size medium, white, polyembryonic, cotyledons whitish, inner coat light brown.

Yield: Good bearer.

Leading Characters: Star Ruby: Introduction from California, an early to midseason, good bearer Grapefruit but sensitive to sun burning which causes blotchy chlorotic leaves, tree medium dwarf spreading, foliage dark green, dense, petiole wings large, fruit medium, light yellow with deep pink blush, rind medium thin, medium loose, medium smooth, oil glands bold, conspicuous, flesh ruby pink, vesicles larger than Ruby, juice high, color dark reddish, flavor rich, sweet, fine, aromatic.

WEBBER SHADDOCK

Type: Pummelo (*Citrus grandisi*); Webber Shaddock, Shaddock, a midseason medium bearer variety of Pummelo grown on small scale.

Origin: Introduction from USA; Origin selection

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Large, moderately vigorous tree, branches spreading wide, medium droopy, water shoots almost spineless, (Rootstalk Rough Lemon).

Leaf: Foliage dark green, medium dense, leaf undulation weak, leaf size large, length 11-12, width 7-8, petiole 1-2 cm, petiole wings very large, aroma very mild, pleasant.

Flower: Flowering season spring, anther yellow, aroma strong, pleasant.

Fruit: Light yellow, size large, round or obovate, depressed on ends, base slightly more depressed than apex, length 9-13, width 12-16 cm, rind thick, medium tight, very rough pebbled, oil glands size bigger, density medium, very conspicuous; albedo pinkish, colomella hollow, segments 14-16.

Flesh and Flavor: Light creamy, fibers hard, vesicles large, opening at sutures, juice medium, color light pink, acidity low 0.15-0.18%, TSS 8-9%, taste good, flavor pleasant.

Seed: Seeds many, 40-50, big, white, monoembryonic, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Webber Shaddock: Introduction from USA, midseason, medium bearer variety of Pummelo, tree large spreading, branches medium droopy, almost spineless, foliage dark green, medium dense, leaf size large, petiole wings large, aroma very mild, pleasant, flowering season spring, fruit large, round or obovate, depressed on ends, color light yellow, rind very thick, medium tight, very rough, strongly pebbled, oil glands size medium, density medium, conspicuous, albedo pinkish, flesh light pink, seeds many, vesicles large, juice medium, color light pink, flavor good, slightly bitter.

LIMES and LEMONS

CHINESE LIME

Type: Limequats (*Citrus aurantifolia*); Eustis, Chinese lemon *Cheeni Nimbu*, high bearer hybrid variety of Limequat (Lime x Kumquat), mostly grown as pot plant and in kitchen gardens on medium scale.

Origin: Introduction; Origin a Limequat hybrid of *Citrus orientifolia* x *Citrus firtunella*).

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Dwarf spreading bushy tree, branching short and spreading, can be grown in pots and most suitable for kitchen gardens.

Leaf: Foliage light green, medium dense, size medium, broadly lanceolate, leaf undulation weak, length 3-5, width 2-3, petiole 0.5-1.0 cm, aroma strong, refreshing.

Flower: All season, aroma strong, pleasant.

Fruit: Medium to light green at maturity, lemon yellow on ripening, size small to medium, roundish, or short elliptical, length 2-3, width 2-4 cm, base rounded, apex round, rind thin, tight, surface smooth, oil glands small and dense, inconspicuous; albedo white, columella solid, segments 8-10.

Flesh and Flavor: Greenish white, fine grained, fibers hard, vesicles small tender, juice high 45-50%, color whitish, acidity high 3-4%, TSS 5-6% aroma strong, distinctive.

Seed: Many seeded 10-20, size medium, white, cotyledons light green, inner coat light brown.

Yield: High yielder.

Leading Characters: Chinese Lime: Introduction, an all season, high yielder hybrid Limequat (Lime x Kumquat), more suitable as pot and kitchen garden plant, tree dwarf, bushy, branches short, spreading, foliage light green, medium dense, flowering and fruiting all the year round, fruit small to medium, roundish or short elliptical, green to light green turning lemon yellow, rind thin, tight, oil glands small, inconspicuous, flesh greenish, juice high, flavor fine, rich, aromatic.

DEPERSE

Type: Acid Lime (*Citrus aurantifolia*); Deperse, Deperse Lime, a midseason variety of Acid Lime.

Origin: Introduction from France;

Maintainer: GPU, Sherkhana.

Tree: Medium large, medium vigorous, bushy branches semi spreading, less spiny.

Leaf: Foliage medium green, medium dense, size medium, broadly lanceolate, leaf undulation weak, length 8-10, width 4-6, petiole 1-2 cm, petiole wings absent, aroma strong, refreshing.

Flower: Flowering mainly spring, anther yellow, aroma strong, pleasant.

Fruit: Green to light green at maturity and lemon yellow on ripening, size small to medium, roundish, or short elliptical, length 5-7, width 4-7 cm, base rounded, apex round, rind thin, tight, surface smooth, oil glands size medium, density medium, conspicuous; albedo white, colomella solid, segments 10-12.

Flesh and Flavor: Light green, fine grained, fibers hard, vesicles medium, hard, juice high 50-55%, color whitish, acidity high 3-4%, TSS 5-6%, aroma strong, distinctive.

Seed: Many seeded 9-10, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Good bearer.

Leading Characters: Deperse: Introduction from France, a midseason, good bearer variety of Acid Lime, tree medium large, bushy, branches semi spreading, foliage medium green, medium dense, petiole wings absent, fruit size medium, roundish to short elliptical, oil glands size medium, density medium, conspicuous, light green at maturity, turns lemon yellow at ripening, rind thin, tight, smooth, oil glands medium conspicuous, flesh light greenish, vesicles medium, hard, juice high, color whitish, acidity high, flavor fine, rich, acid, aromatic.

EUREKA LEMON

Type: Lime (*Citrus limon*); Eurika Lemon, a midseason, good bearer, medium scale variety, bearing solitary and terminal clusters.

Origin: Introduction from California 1973; Origin seedling or nucellar clonal selections of Italian material.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium to tall, medium vigorous, open, long semi spreading branches, water shoots virtually thornless, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium lax, leaf undulation weak, length 10-12, width 5-7, petiole 1-2 cm, wingless, aroma strong, refreshing.

Flower: Solitary and terminal clusters, flowering throughout the year but mainly spring, anther yellow, fragrance medium, pleasant.

Fruit: Light green at maturity, lemon yellow at ripening, size medium to large, elliptical to oblong, length 9-10, width 6-8 cm, base round with short neck, apex with medium nipple usually with areolar furrow, finely pitted, medium to strong longitudinal ridges, rind medium thick, tight, medium rough, oil glands tiny, numerous, sunken; albedo white, colomella medium, solid, segments 10-12.

Flesh and Flavor: Greenish yellow, fibers hard, vesicles medium soft, tender, juice high 50-52%, fine grained, color whitish, acidity high 5-6%, TSS 7-8%, aroma medium, pleasant, highly acidic.

Seed: Few seeded 1-10, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: High bearer.

Leading Characters: Eureka Lemon: Introduction from California, high bearer, tree medium to tall, foliage medium green, medium lax; fruit solitary and terminal clusters, elliptical to oblong, light green at maturity, pale yellow at ripening, longitudinal ridges prominent, rind medium thick, tight, flesh greenish yellow, juice high, good for flavoring, aromatic, (strong ridges, more rough rind, a smaller nipple, more round as compared to Lisbon).

FROST LISBON

Type: Lemon (*Citrus limon*); Frost Lisbon; medium maturing, medium yielding, variety lemon.

Origin: Introduction from France; Origin nuceller selection of Lisbon by H. B. Frost, at Florida, 1917.

Maintainer: GPU, Sherkhana.

Tree: Medium to large, medium vigorous, upright, open, water shoots medium thorny, (Rootstalk **Rough Lemon**).

Leaf: Foliage medium green, medium dense, leaf undulation weak, large, length 10-12, width 5-7, petiole 1-2 cm, wingless, aroma strong, refreshing.

Flower: Flowering throughout mainly spring and autumn, solitary and terminal clusters, anther yellow, fragrance medium, pleasant.

Fruit: Light green at maturity, lemon yellow at ripening, size medium, elliptical to oblong, length 7-9, width 6-8 cm, base round to tapering with short neck, apex with medium nipple, furrowed, finely pitted, medium longitudinal ridges, rind medium thick, tight, medium rough, oil glands size tiny, density numerous, sunken, less conspicuous, albedo white, colomella small, solid, segments 10-12.

Flesh and Flavor: Greenish yellow, fibers hard, vesicles medium soft, tender, juice high 45-50%, fine grained, color whitish, acidity high 4-5%, TSS 7-9%, aroma medium, pleasant, acidity high.

Seed: Few seeded 0-7, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium bearer.

Leading Characters: Frost Lisbon: Introduction from France, originally nuceller selection of Lisbon, tree medium large, upright, foliage medium green, medium dense, fruit size medium, elliptical to oblong, light green at maturity, lemon yellow at ripening, rind medium thick, tight, oil glands numerous, inconspicuous, flesh greenish yellow, juice high, good for flavoring, aromatic.

GALGAL

Type: Rough Lemon (*Citrus pseudolimoni*); (*Gagal*, *Gulgul*), early season good bearer variety.

Origin: Local selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium to tall, vigorous, bushy, upright to spreading, water shoots thorny, thorns numerous and thick, propagation by seed.

Leaf: Foliage medium to light rather dull green, medium dense, size large, medium cupping. undulation weak, medium broad, length 8-10, width 4-5, petiole 1-2 cm, petiole wings large, aroma medium, refreshing.

Flower: Flowering main season spring, purple tinged, anther yellow, aroma strong, pleasant.

Fruit: Pale to golden yellow, size medium to large, oblong to ellipsoid, length 12-15, width 10-12 cm, base round tapering, medium furrowed neck, apex depressed, flat or with short to blunt or pointed nipple, rind thick, medium tight, easy peel, rough, oil glands small, numerous, medium conspicuous; albedo white, colomella large, semi hollow, segments 10-14.

Flesh and Flavor: Pale yellow, coarse fibers strong, vesicles large, tight, juice medium 33-35%, color whitish, acidity high 5-6%, TSS 7-8%, taste bitter, aromatic modernly acidic.

Seed: Many seeded 15-25, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: High bearer.

Leading Characters: Galgal: Local selection of unknown origin, good bearer, tree medium to tall, vigorous, upright to spreading, foliage medium dull green, flowering in spring, purple tinged, fruit medium to large, neck medium furrowed, apex depressed, flat or with blunt to pointed nipple, pale to golden yellow, rind thick, medium tight, easy peel, smooth to medium rough, flesh pale yellow, juice medium, whitish, flavor bitter, aromatic.

KAGHZI LIME

Type: Acid Lime (*Citrus aurantifolia*); Kaghazi Lime, *Kaghzi Nimbu*, West Indian Lime, Key Lime or Mexican Lime, a good bearer popular variety grown on large scale.

Origin: Local selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium large, spreading bushy, branches short to medium long, semi spreading to spreading, highly spiny, spines medium short.

Leaf: Foliage medium green, dense, leaf size medium, broadly lanceolate, undulation weak, length 4-6, width 3-4, petiole 1.0 cm, petiole wings short, aroma strong, refreshing.

Flower: Flowering all season, mainly spring, anther yellow, aroma strong, pleasant. Young flowers faintly purple tinged.

Fruit: Green to light green, lemon yellow on ripening, small to medium, roundish, or short elliptical, length 2-3, width 3-4 cm, base rounded occasionally short to medium necked, apex round, slight to medium furrowed nipple, rind thin, tight, surface smooth, oil glands small, dense, conspicuous; albedo white, colomella solid, segments 10-12.

Flesh and Flavor: Greenish, yellow, fine grained, fibers hard, vesicles small tender, juice high 50-55%, color whitish, acidity high 3-5%, TSS 5-6%, aroma strong, distinctive.

Seed: Many seeded 8-15, medium, white, polyembryonic, cotyledons light green, inner coat light brown. Due to high degree of polyembryony it comes remarkably true to seed hence generally seed propagated.

Yield: High yielder.

Leading Characters: Kaghzi Lime: Local selection, high yielder acid lime, tree medium large, bushy spreading, foliage medium green, flowering and fruiting all the year round, young flowers faintly purple tinged, fruit medium (smaller than Tahiti), green to lemon yellow, slight to medium furrowed nipple, rind thin, tight, flesh greenish yellow, juice high, fine rich aromatic.

LISBON

Type: Lemon (*Citrus limon*); Lisbon, a midseason, medium bearer, medium scale commercial variety.

Origin: Introduction from California 1973; Origin Portuguese brought to California.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium to tall, medium vigorous, upright, open, water shoots medium thorny, (Rootstalk Rough Lemon).

Leaf: Foliage medium green, medium lax, leaf undulation weak, length 10-12, width 5-7, petiole 1-2 cm, petiole wings absent, aroma strong.

Flower: Flowering more the once, mainly spring and autumn, solitary and terminal clusters, anther yellow, fragrance medium, pleasant.

Fruit: Light green at maturity, lemon yellow at ripening, size medium to large, elliptical to oblong, length 9-12, width 6-9 cm, base round to tapering with short neck, apex with prominent nipple usually with areolar furrow, finely pitted, weak longitudinal ridges, rind medium thick, tight, medium rough, oil glands tiny, numerous, sunken; albedo white, colomella small, solid, segments 10-12.

Flesh and Flavor: Greenish yellow, fibers hard, vesicles medium soft, tender, juice high 48-50%, fine grained, color whitish, acidity high 4-5%, TSS 7-8%, aroma medium, pleasant, acidity high.

Seed: Few seeded 0-7, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Good bearer.

Leading Characters: Lisbon: Introduction from California, a midseason, medium bearer, tree medium to tall, foliage medium green, medium lax; fruit size medium, elliptical to oblong, color light green at maturity, longitudinal ridges less prominent, rind medium thick, flesh greenish yellow, juice high, good for flavoring, aromatic, (Lisbon smoother, less ridged, nipple more prominent, leaves less dark and less crenate than Eureka).

MEXICAINE

Type: Acid Lime (*Citrus aurantifolia*); Mexicaine Lime, midseason, medium bearer variety of Acid Lime.

Origin: Introduction from France; Origin selection.

Maintainer: GPU, Sherkhana.

Tree: Medium large, medium vigorous, upright, branches upright to semi spreading, about spineless.

Leaf: Foliage dark green, medium dense, leaf size large, broadly lanceolate, leaf undulation weak, length 12-14, width 7-9, petiole 1-2 cm, petiole wings absent, aroma medium, refreshing.

Flower: season spring, anther yellow, aroma strong, pleasant.

Fruit: Green to light green at maturity, lemon yellow on ripening, size small to medium, ovoid or short elliptical, length 5-7, width 4-8 cm, base rounded, apex rounded, rind thin, tight, surface rough, oil glands size medium, density medium, medium conspicuous; albedo white, colomella solid, segments 10-12.

Flesh and Flavor: Light green, fine grained, fibers hard, vesicles medium, hard, juice high 50-55%, color whitish, acidity high 3-4%, TSS 5-6%, aroma strong, distinctive.

Seed: Many seeded 5-15, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium yielder.

Leading Characters: Mexicaine: Introduction from France, a midseason variety of Acid Lime, tree medium large, upright, branches upright to semi spreading, foliage dark green, medium dense, leaf size large, petiole wings absent, flowering spring, fruit size small to medium, ovoid to short elliptical, oil glands size medium, density medium, conspicuous, fruit color green to light green at maturity, turning lemon yellow at ripening, rind thin, tight, smooth, oil glands medium conspicuous, flesh light greenish, juice high, color whitish, acidity high, flavor fine, rich, aromatic.

ROUGH LEMON

Type: Rough Lemon Sour Lime (*Citrus jambhiri*); *Jatti Khatti*, *Jamberi*, most extensively used rootstalk for citrus fruits in Pakistan and India. It can be easily propagated by cuttings but generally propagated by seed, seedlings 90-100% of nucellar origin, considered true-to-type.

Origin: Local selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium to tall, vigorous, bushy, well branched, upright spreading, water shoots small to medium thorny, (Rootstalk Rough Lemon).

Leaf: Pale to medium green, medium dense, medium small, blunt, undulation weak, long, medium broad, length 8-10, width 4-5, petiole 1-2 cm, petiole wings narrow, aroma medium, flush anthocyanin medium to strong.

Flower: Flowering mainly spring but casual flowers in autumn, purple tinged, anther yellow, aroma strong, pleasant.

Fruit: Light green later turning lemon yellow to orange or lemon brown, size medium, oblate to elliptic oblong, length 6-8, width 5-7 cm, base round tapering, weak to medium neck, apex round with medium to large naval surrounded by deep irregular furrow, rind thick, medium tight, bumpy, sometimes ribbed, oil glands numerous, medium conspicuous; albedo white, colomella semi hollow, segments 10-12.

Flesh and Flavor: Light yellow, fibers strong, vesicles large, tight, juice medium 35-37%, color whitish, acidity high 4-5%, TSS 4-5%, j taste bitter, aromatic moderately acidic.

Seed: Many seeded 15-25, size small to medium, creamy white, polyembryonic, cotyledons creamy white, inner coat light brown.

Yield: High bearer.

Leading Characters: Rough Lemon: Local variety, high bearer, tree medium to tall, spreading, foliage medium green, fruit medium, rind thick, bumpy, rough, flesh white, juice medium, white, aromatic, most extensively used rootstalk in Pakistan and India.

SWEET LIME

Type: Acidless Lime, Sweet Lime (*Citrus limettoids*); Sweet Lime, *Mitha*, Palestine lime, Indian lime, midseason, high bearer, popular commercial variety.

Origin: Local selection.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium tall, spreading medium, branches semi spreading, water shoots thick thorny.

Leaf: Foliage medium to pale green, medium dense, leaf medium rolled, undulation weak, length 7-9, width 3-4, petiole 0.5-1.0 cm, petiole wings absent, aroma strong, refreshing.

Flower: Flowering spring, anther yellow, fragrance distinctive, pleasant.

Fruit: Light green at maturity (consumption stage), turns yellow on ripening, size medium, sub globose to oblong or short elliptic, length 6-7, width 5-6 cm, base and apex round, rind thin to very thin, tight, surface smooth sometimes faintly ribbed, glossy, oil glands small, numerous, prominent; albedo white, colomella medium, semi hollow, segments 10-12, segmental depressions visible.

Flesh and Flavor: Straw yellow, fibers hard, vesicles medium tender, juice high 45-50%, whitish, acidity very low 0.2-0.3%, TSS 8-9%, flavor insipid, sweet slightly bitter after taste, aroma of rind distinctive, pleasant.

Seed: Many seeded 10-20, size medium, white, polyembryonic, cotyledons light green, inner coat light brown, chalazal spot light tan.

Yield: Good bearer.

Leading Characters: Sweet Lime: Local selection, high yielding, tree medium tall spreading, foliage medium green, fruit medium, light green to pale yellow, flesh straw whitish, rind thin, tight, juice high, taste good, aromatic, considered to have special medicinal value in the prevention of malaria fever and as a friend of liver. Propagation mostly by rooted-cutting nursery trees instead of seedling trees.

TAHITI

Type: Lime (*Citrus latifolia*); Tahiti, Tahiti Lime, Persian lime, a medium bearer variety grown on medium scale.

Origin: Introduction from California 1973; Origin unknown.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Medium large spreading, vigorous, bushy with drooping branches, highly spiny, spines medium to long, (Rootstalk Rough Lemon).

Leaf: Foliage dark to medium green, medium dense, leaf size medium, leaf undulation absent to weak, length 4-8, width 4-5, petiole 0.5-1.0 cm, petiole wings sort narrow, aroma strong, refreshing.

Flower: All season, maximum in spring, anthers yellow, aroma strong, pleasant.

Fruit: Green to light green and lemon yellow on ripening, size medium, ovate to oblong, length 5-8, width 4-7 cm, base round, neck slight with slight furrows, apex round, nipple short, medium furrows, rind thin, tight, surface smooth, oil glands small and sparse, conspicuous; albedo white, colomella small, solid, segments 10-12.

Flesh and Flavor: Pale greenish yellow, fibers hard, vesicles small, tight, tender juice high 42-45%, color whitish, acidity high 4-5%, TSS 7-8%, juice color whitish, flavor pleasant true lime flavor.

Seed: virtually seedless 0-5, size medium, white, polyembryonic, cotyledons light green, inner coat light brown.

Yield: Medium yielder.

Leading Characters: Tahiti Lime: Introduction from California, medium bearer lime, tree medium, vigorous, bushy spreading, branches droopy, foliage dark to medium green, size medium, (leaf bigger than Kaghzi lime), flowering and fruiting all the year round (maximum in spring), fruit medium, green to lemon yellow at maturity, nipple short, rind thin (thicker than Kaghzi lime), tight, surface smooth, flesh pale greenish, juice high (less than Kaghzi lime), color whitish, fine rich aroma (less than Kaghzi lime), flavor pleasant, true lime flavor

Citrus varieties

Kumquats

KUMQUATS

MEIWA

Type: Kumquat (*Fortunella margarita*); Meiwa, *Maiwa*, or Large Round Kumquat, a common ornamental variety of Kumquat, mostly grown as pot plant and home gardens.

Origin: Introduction; Origin clonal selection of Ninpo or Meiwa or Neiha kinkan of Japan.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Dwarf upright semi-bushy tree, branching short and upright, medium spiny, grown in pots and most suitable for home gardens.

Leaf: Foliage light green, medium dense, size medium, broadly lanceolate, leaf undulation weak, length 7-8, width 2-3, petiole 1.0 cm, petiole wings very small, aroma refreshing.

Flower: All season, mainly spring, flowers appear in bunches, aroma strong, pleasant.

Fruit: Dark orange attractive, size small to medium small, shape oblong, obovate, or oval, length 2-2.5, width 2.5-3.0 cm, base rounded, apex round, rind thick, sweet, tight, surface smooth, oil glands small, dense, inconspicuous; albedo white, columella hollow, segments 4-7.

Flesh and Flavor: Creamy white, fine grained, fibers hard, vesicles small tender, melting, juice high, color creamy, acidity low, eaten as a whole with skin, flavor good, aroma strong, distinctive.

Seed: Few seeded 2-4, size medium, white, cotyledons light green, inner coat light brown.

Yield: Good bearer.

Leading Characters: Meiwa: Introduction, a popular, good bearer horticultural Kumquat, tree dwarf, bushy spreading, medium spiny, spines smaller than Nagami, foliage light green, flowering and fruiting all the year round, fruit small to medium relatively larger than Nagami, dark orange, rind thin, tight, smooth, flesh creamy, juice high, flavor fine, rich, comparatively sweeter than Nagami, aromatic, eaten whole with skin.

NAGAMI

Type: Kumquat (*Fortunella margarita*); Nagami, *Naghmi*, Oval Kumquat, a common ornamental variety of Kumquat, mostly grown as pot plant and home gardens.

Origin: Introduction; Origin clonal selection of Naga or Nagami kinkan of Japan.

Maintainer: GPU, Sherkhana; ORI, Sargodha.

Tree: Dwarf spreading upright semi-bushy tree, branching short and upright, medium spiny, can be grown in pots and most suitable for home gardens.

Leaf: Foliage light green, medium dense, size medium, broadly lanceolate, leaf undulation weak, length 7-8, width 3-4, petiole 1.0 cm, petiole wings very small, aroma refreshing.

Flower: All season, mainly spring, flowers appear in bunches, aroma strong, pleasant.

Fruit: Dark orange beautiful color, size small to medium, shape oblong, obovate, or oval, length 2-3, width 3-4 cm, base rounded, apex round, rind thin, tight, surface smooth, oil glands small, dense, inconspicuous; albedo white, columella hollow, segments 4-6.

Flesh and Flavor: Creamy yellowish, fine grained, fibers soft, vesicles small tender, melting, juice high, color whitish, acidity low, eaten as a whole with skin, flavor good, aroma strong, distinctive.

Seed: Few seeded 2-5, size medium, white, cotyledons light green, inner coat light brown.

Yield: High yielder.

Leading Characters: Nagami: Introduction, a very popular good bearer horticultural Kumquat, tree dwarf, bushy spreading, medium spiny, spines bigger than Miwa, foliage light green, flowering and fruiting all the year round, fruit small to medium, dark orange, rind thin, tight, smooth, flesh yellowish creamy, juice high, flavor fine, rich, sweet, aromatic, eaten whole with skin, a bit sour than Meiwa.

KINDS OF CITRUS FRUITS



Citrus fruits are perhaps the most popular and extensively grown fruits occupying third place for cultivated area on world level after grapes and olives. But they are comparatively recent in development as compared to other fruit of the world. Their original homeland is considered to be southern and southeastern mainland of Asia and Malayan islands. The earliest cultivation of citrus is from China about 2200 BC. In Pakidtan the most extensively grown species is *C. reticulata* (Kino), however there are many other species produced on various levels like:

Sr.No.	Local name	Botanical name	Common name
1.	Chakotra	<i>Citrus grandis</i>	Pummelo, shaddock
2.	Galgal (large), Mokari	<i>Citrus medica</i>	Citron
3.	Grapefruit	<i>Citrus paradise</i>	Grapefruit, pomelo
4.	Jatti khatti, Jamberi	<i>Citrus jamberi</i>	Rough lemon
5.	Kharna khatta	<i>Citrus kerna</i>	Kerna lime
6.	Khatta	<i>Citrus aurantium</i>	Sour orange
7.	Kino, Sangtra	<i>Citrus reticulata</i>	Mandarin
8.	Malta	<i>Citrus sinensis</i>	Sweet orange
9.	Mitha	<i>Citrus limettoides</i>	Sweet lime
10.	Kaghzi Nimboo	<i>Citrus aurantifolia</i>	Acid lime
11.	Nimbu	<i>Citrus limonia</i>	Lemon, Lisbon Lemon

Citrus fruits are prized for their health values. oils, pectin, flavorings, perfumes and other by-products are secured from their flowers and fruits.

From the commercial point of view on international level only eight species are most important: Citron, *Galgol* (*Citrus medica*)

It grows as an irregular, open-headed shrub or small tree with large, light green leaves. The flowers are painted purple on the outside. The fruit size is large, shape oblong or ellipsoidal. The peel is very thick and rough, yellow on the outside and white inside. The pulp is greenish with very low and very acidic juice in most types. Its main use is of candying the peel and candied peel is the main Citron product.

Grapefruit (*C. paradisi*)

The grapefruit tree is vigorous growing up to 30 to 50 feet tall and has a round, thick head of foliage. Its leaves are large, ovate and blunt pointed. The flowers are large, white and produced singly or in axillary clusters. The fruits are larger than the Orange and are usually in bunches and clusters like grapes. The “grape”-like bunches are the cause of the name “grapefruit”. The fruit color is light yellow having silvery white or pink flesh. The white seeds are larger than others. The flavor is a blend of acid, sweet and bitter.

Lemon, *Nimboo* (*C. Limonia*)

The lemon tree is small and widely branched growing up to 10 to 20 feet high. It is thorny and evergreen and its leaves are narrow and ovate and light green. The flower buds are in pairs or single with petals white inside and purple on the outer surface. The fruit is usually pointed at both ends and light yellow. The flesh is light and seeds ovoid. The Lemon is grown for its acid juice, which is used in flavoring and in making various drinks. Lemon peel is candied. The origin of the Lemon is unclear.

Lime, *Kaghzi Nimboo* (*C. aurantifolia*)

The lime tree is evergreen, small, spiny and irregularly branched. It has small, elliptic to oblong pale green leaves. The white flowers are small and produced in axillary clusters. The fruit is small, roundish and thin-skinned. The pulp is greenish with distinctive flavored acidic juice. The Lime is a native of the East Indies and has spread all over the world in tropical and near tropical regions. The fruit is used in the same way as Lemon.

Mandarin Orange, Santra, Sangtra, Kino (*C. reticulata*)

This small, spiny tree has thin branches and twigs. Its leaves are small and elliptic to lanceolate. The flowers are small and white and produced singly or in small clusters. The oblate (often very flattened at the ends) fruits are orange to orange-red with a loose rind that comes off easily. The 9-15 fruit sections are loosely attached to the rind and one another. The pulp is deep orange, sweet or fairly acid. The green seeds are numerous. This tree is native to southeastern Asia. The fruit yields a high quality juice and is used as a desert fruit. Mandarins are a large distinctive and highly varied group that includes some of the finest and most highly reputed citrus fruits like loose skin oranges-mandarin or Tangerines including santra or sangtra. Some of the distinctive features follow:

- Rind loose, Sections loosely adherent, open core, flavor and area distinctive seed with greenish cotyledons.
- Distinctive leaf petiole-wings.
- Blade notch-pointed and main view prominent above as well as below.
- Spines small and few or absent
- Flowers single or in clusters.

Shaddock (*C. maxima*)

This large tree has a round top with fuzzy twigs when young. Its huge leaves are ovate. The flowers are white, borne singly or in bunches. The fruit is the largest of all citrus. It is globose, oblate or moderately pear-shaped with 11-14 sections. The pulp is light colored or pink and coarse with large, spindle-shaped juice sacks that separate easily from one another. This tree is a native of southeastern China. It was Introduced by Captain Shaddock hence the name Shaddock. Pummelo is another name for it.

Sour Orange, Khatta (*C. aurantium*)

This small, spiny tree grows up to 30 feet high and has a compact, rounded top. Its leaves are ovate and fragrant. The fruit is orange or reddish-orange with rough, strong scented and bitter rind. There are 10-12 fruit sections and the pulp is acid. The Sour Orange is grown extensively

in some Citrus regions for its fruit, which is used in making marmalade. In areas where it grows, it's used for making acid fruit drinks that are very agreeable in warm weather. It is much hardier than the Sweet Orange.

Sweet Orange, *Malta* (*C. sinensis*)

The tree is medium-sized (25-40 ft.) with a compact round-oblong head. The leaves are ovate to ovate-oblong. The flowers are white, fragrant borne in small clusters on the new growth. The fruit is round or ovoid and colored orange to reddish-orange. The juicy flesh is orange and fairly acid with few or many seeds. This is native to the region comprised of Northeastern of the Subcontinent, and adjoining portions of China and Burma and was known as *Malta*. Sweet Oranges are grown extensively in all parts of the world. The sweet oranges fall into four kinds.

1. Common oranges
2. Acidless oranges
3. Pigmented oranges
4. Navel oranges.

Common Oranges

Mosmbi, Jaffa, Valencia late, Salustiana.

Sugar or Acidless Oranges

This group comprise of a small group of very low acid varieties like sweet lamons, limettas, and limes. The chalazal spot on the seeds of these acid lacking varieties is cream-colored, whereas darker on more acid forms, and chestnut brown on other common sweet oranges.

Blood Oranges

Blood or pigmented oranges appear to have originated in the Mediterranean basin. They differ in appearance from common sweet oranges only because under certain conditions the fruit usually exhibits pink or red coloration in the flesh, juice and on the rind. They are characterized by:

- Distinctive flavor that adds to the deliciousness of the fruit.
- The blood coloration which is due to the development of anthocyanin pigments whereas the pink and red coloration in

the pigmented grapefruit is caused principally by the cerotenoid pigment lycopene. These pigments tend to deteriorate during processing and impart a muddy color to juice.

- Color development is variable under different environmental conditions. Cold storage intensifies flesh coloration.
- Rind coloration is usually best on less exposed or partially shaded fruits.
- Blood coloration may vary in response to nutrition, changes in environmental conditions, and rootstock.
- Torocco, Moro and Sanguinelli exhibit purplish red coloration of the chaazal spot on seed whereas other varieties have brown spot.
- The uniformity of coloration of flesh and juice is unappreciable.

There are some cultural practices for the development of coloration like harvesting is said to consist of picking the north and south sides of the trees separately, the northern side more pigmented. In Pakistan some farmers shade the lower portion of trees by means of a tall rapid growing cover crop such as sesbania (*Sesbania sesbane*).

Navel Oranges

Consists of presence of a navel which is a small and rudimentary secondary fruit embed in the apex of the primary fruit. The tendency to produce fruits with small navel is exhibited by a number of the mandarins and some other orange varieties.

- It is highly variable from season to season.
- Only in the navel oranges does this phenomenon occur regularly.
- Seedlessness is characteristic of navel oranges, resulting from the fact the functional pollen is lacking and viable ovules are rare.
- Crispiness of flesh, ease of peeling, separation of segments and richness of flavor and deep orange color are other characteristics.

Hybrids

Most of the Citrus hybridize quite easily and a large number of hybrids have been created. Many of these aren't superior to the originals, but others are worth while. The Tangelo is a hybrid between the Grapefruit and Tangerine. It is one of the best of the hybrid Citrus fruits and some varieties are grown commercially. Another one is the Citrange. It is a hybrid between the Sweet Orange and the Trifoliate Orange (*Poncirus trifoliata*). There are many other hybrid citrus fruits but none of the varieties have fruits suitable for eating out of hand, however, their main value is for making preserves.

Kumquat

The Kumquat (*Fortunella*) and Trifoliate Orange (*Poncirus*), mentioned above and both less important, are also included under the common name Citrus, but don't belong in the botanical group Citrus.

CITRUS TREE

Citrus trees are small, spiny shrubs or trees with alternate, evergreen, shiny leaves having oil glands, small pleasant smelling flowers occurring singly or in clusters, fruit a berry or egg shaped hesperidium with thick leathery peel, colored exocarp having embedded oil glands, usually 8-14 juicy sections, each section having none to many seeds. A brief general description follows:

Stem

The stem is mostly a single trunk with many main branches. Spines occur in some species and are adjacent to buds in leaf axils. New growth of shoots occurs in definite cycles usually three every year. The buds present on the main stem may remain dormant for many years or may grow out as water-shoots, particularly when top of tree is damaged.

Leaf

The leaf is unifoliate, evergreen, leathery, shining, wings present on petiole in some species.

Flowers

The flowers singly or in small clusters smell sweet and they have five petals that are white and some kinds have purple staining the outer surfaces, usually hermaphrodite. Most species are self- cross-compatible and are entomophilous visited by bees which are attracted by white corolla, strong perfume abundant nectar and sticky pollen.

Fruit

The fruit a berry or hesperidium is spherical or egg-shaped and usually have 8-14 juicy sections. The fruit has thick leathery peel; exocarp or flavedo is colored outer portion with oil glands embedded in tissues. Albedo or mezocarp is inner colorless section of peel rich in sugars, pectin, vitamin C and glucosides. In naval oranges there is partial development of a second whorl of carpels above first which produces umbilicus effect. Seed a dicot with hypogeal germination.

Seed

Number of seed per fruit and per locule varies greatly from seedless to many. Most cultivars except *C. grandis* have polyembryonic seeds which may consist of sexual or gametic embryo from fertilized egg cell and up to 6 or more asexual or nucellar embryos, which develop from somatic cells of nucellus and therefore have the same genetic constitution as female parent. Gametic embryo often lacks vigor and may be completely suppressed by nucellar embryos.

CITRUS SEED PRODUCTION



Citrus rootstocks are normally propagated through seed. In view of the influence of rootstock on longevity, productivity, fruit quality, yield and horticultural value, it is very important that the rootstock should be most appropriate and capable of producing sufficient quantities of good seed. One of the most important factors in citrus is the presence of polyembryony in the seed. Majority of the citrus varieties have polyembryonic seeds except Chakotra or Pummelo (*Citrus garandis*). The polyembryonic seeds may consist of sexual or gametic embryo from fertilized egg cell and up to six or more asexual or nucellar embryos, which develop from somatic cells of nucellus and therefore have the same genetic constitution as the female parent. The gametic embryos normally lack vigor and may be completely suppressed by nucellar embryos. This feature offers a great help to nurserymen for selecting vigorous nucellar seedlings.

Seed Stock

For producing high quality, healthy and disease free citrus nursery trees it is necessary that the rootstock as well as scion should be healthy and disease free. To obtain healthy and uniform quality seed, the seed source trees must meet the following criteria:

1. The seed tree represents the recommended rootstock variety, is genetically true-to-type with normal shape and morphological characteristics.
2. The seed tree is vigorous, healthy and free of bark and leaf symptoms of harmful virus diseases, and other serious pathogens.
3. The seed tree is high yielding and the fruit quality meets the desirable horticultural characteristics.

4. The seed tree is genetically stable for the shape, size and quality of fruit, which should be free of excessive deformation and mutations.
5. For producing high quality certified nursery trees by the registered nurseries, the seed tree should be visually examined for true-to-type morphological characteristics, indexed for freedom from prescribed diseases and other pathogens and registered by the Federal Seed Certification and Registration Department.

Fruit Collection

Collection of fruit for seed extraction from reliable sources minimizes the chances of seedling variation. The fruit selected for seed extraction should possess good fruit and keeping quality characteristics. The fruit collected for seed should be tree-ripe, healthy and sound. An approximate estimate of requirement of Rough Lemon (*C. jambhiri*) fruit for extracting seed should be three to four times the number of seeds required to account for failures, selection and rejection. For making a rough estimate of the requirement of ripe fruit for extraction of certain amount of seed, the calculations may be based on the presumption that the average yield of seed from Rough Lemon, Sweet Lemon, Sour Orange and Trifoliate Orange has been estimated to range between 20-25 seeds/fruit. The pummelo can yield 50-60 seeds/fruit. The fruit normally is collected in September.

Seed Extraction

The seed is extracted manually by hand by cutting the fruits carefully in to halves and then twisting to squeeze out the seeds. It is a good practice to squeeze the seed into a sieve placed over a trough. The seed must be washed free of pulp and juice. Seeds must be immersed for 10 minutes in well agitated warm water to get off gelatinous material. While washing, the good and healthy seed will sink to the bottom whereas the immature and poorly developed seeds will float. Remove surface moisture with absorbent cloth or paper and dry them in partial shade in an airy place and for best germination plant fresh.

Seed Storage

The storage of citrus rootstock seed under ordinary conditions is not generally recommended because they lose viability and germination very quickly. The citrus seed is very sensitive to drying. If the seeds are over dried, the cotyledons also lose moisture and separate from each other resulting in deterioration of viability and vigor very severely. However, fresh seeds can be stored for short periods under controlled conditions preferably in vapor proof containers. For this purpose the seeds should be washed after extraction and dried off quickly after washing. Then the seed must be mixed with an equal volume of finely ground charcoal or wood ashes and canned. Ordinarily citrus seed lose viability and germination if stored for 2-3 weeks at room temperature. For short storage it is preferred to store the fruit instead of extracted seed. Citrus seed treated with some suitable fungicide like Arasan can be stored in vapor proof containers or in sealed polythene bags at 3-10°C for up to eight months.

Seed Viability

Citrus seed gives highest germination if planted shortly after extraction otherwise the deterioration starts after 48 hours. If the citrus seed from storage has to be used, it will be useful to test the viability prior to sowing. For reliability the nearest seed testing laboratory of the Federal Seed Certification and Registration Department may be contacted to help test the samples for viability and germination. However, it can be tested at the farm by carefully removing the seed coat without injuring the interior and placing the seeds in wet Petri dish, china plate, towel, paper or sand at room temperature or in an incubator at 25°C for 3 days. The towel, paper or other media used must be kept moist by sprinkling fresh water one to two times daily during this period otherwise under dry conditions the seed will not germinate. In urgent situations the seed viability may be tested by quick tetrazolium test. In this method the citrus seeds are cut longitudinally in the laboratory and placed in 1% solution of 2,3,5-triphenyl tetrazolium chloride (TTC) for 24 hours. The viable seeds will turn bright pink.

Seed Sowing

General recommendation and common practice of sowing Rough Lemon and other rootstock seed is in September. However, satisfactory

germination and good vigorous growth of seedlings have been obtained from mid February-March sown stocks. For mid February sowing the farmer must take care that there is no danger of young seedlings being damaged by frost, for if they are frozen they are killed.

To reduce danger of infestation of diseases and pests, the site selected for sowing must preferably be away from citrus orchards and old citrus trees and should have not been used for nursery raising during the previous year. Seedlings can be grown on a variety of media like flat soil, raised seedbeds, containers or trays. To ensure freedom from diseases and soil pests the seedlings must be raised on clean and disinfected medium. The soil should be thoroughly worked, pulverized and mixed with well rotten FYM or leaf mould. However in general it is preferred to sow seed on well prepared raised seedbeds. For better and quick growth it is preferred to lay out the seedbeds under tunnel conditions with efficient drainage system. Suitable shelters must be provided to protect the young seedlings from frost or heat.

The seed is dibbled 1½ to 2 cm deep at 2-3 cm distance in rows 15 cm apart and covered with sand, leaf mould or soft soil. The seedbeds are sprinkled immediately after sowing and kept moist by sprinkling water daily or when needed. For better phyto-sanitary care the seed can be sown in containers of suitable size or preferably in polythene bags and placed under tunnel conditions. The seed will take 3-4 weeks to sprout.

Seedlings

When the seedlings are about two months old, the weaker and abnormally growing seedlings can be thinned out. As the seed is polyembryonic containing both sexual and asexual seedling. Therefore, while propagating by seed, the sexual seedlings which are usually stunted and poor are rogued out. The rest that are produced from the cells of the nucellus are more vigorous and allowed to grow. The seedlings, thus selected, are more or less uniform in growth and production.

Transplanting

The seedlings are ready for transplanting when they are 4-6 (or more up to 12) months old. The recommended time of transplanting is February-March and September-October. Uniform and vigorous seedlings are transplanted in rows at alternate distance of 25 and 60 cm apart and 25 cm

from plant to plant. The seedlings during transplanting must be kept moist and should not be exposed to direct sun heat. For better growth the seedlings should be given light doses of fertilizer and frequent irrigations and kept free of weeds. The undersized, or plants with abnormal growth, disease symptoms or off-type plants should be rogued out. One year after transplanting, when the diameter of the rootstock stems reaches about 2 cm, they are ready for budding.

CITRUS FRUIT PRODUCTION



Citrus originated in Southeast Asia and Indo-Pak subcontinent, where it is today one of the most important fruit crops. It is grown in more than 100 countries having tropical or subtropical climates. In Pakistan and other Asian countries on an average the farm holdings are small, production and marketing costs are high resulting in low average yields compared to those of western countries. In Pakistan, the average yields of citrus fruit crops are estimated to be only one-third to one-fifth of the genetic potentials. The low orchard efficiency is resulting in below optimum yields and quality compared to USA. New production technology can help farmers reduce production costs and improve the quality of fruit.

The globalization resulting from the WTO agreement ushers in a new era of global free trade, which makes it imperative for Pakistan and other Asian countries to become more competitive in fruit production and processing. This will enable our farmers to hold their place in domestic markets in the face of trade liberalization, and if possible to expand it through exports.

Citrus production suffers from very serious problems of severe infestation of a large number of diseases, insects, pests and weeds. The attack of virus diseases and pests cause the life span of trees to be relatively short and drastically reducing the fruit bearing capacity of the trees. In view of the long time it takes citrus trees to reach fruit bearing stage, increasing the productive life span of trees would make an important contribution to increasing the productivity of the orchards.

The yields can not be improved unless three most basic problems are addressed which include a lack of disease-free genuine nursery trees, a lack of appropriate production package covering input management and cultural techniques and a lack of high yielding improved varieties. Input requirements and cultural techniques will vary depending on the climatic environments, soil conditions and variety type. However general procedures regarding propagation and production follow:

Climate and Soil

Citrus fruits can be produced on a variety of soils from light sandy kinds to clayey, heavy and stiff kinds provided they are well drained and free of alkali. The soils considered best for citrus production are alluvial and loamy soils that hold moisture, has adequate drainage and full of vegetable matter. The heavy soils will need suitable cultural practices and the light soils will have to be supplied with optimum doses of fertilizer and irrigation. Citrus flourish very well in soil that contains high amount of vegetable matter. The green manuring is the best source of adding vegetative matter in the soil and can be easily achieved by growing any suitable leguminous crop like sesbania, *junter*, sun hemp, or berseem and ploughing the green crop to turn the soil under. Most citrus grows well in a soil pH range from 6 to 8. Avoid soils that have high alkali content or are excessively salty, as citrus trees will not grow well in such soils.

The citrus plant thrives best in mild and sub-tropical climates. They can however be grown in warm temperate regions, where minimum temperature in winter does not fall below zero for prolonged periods. Cool weather reduces vegetative growth, makes unripe fruit to loose color and increase acidity. On the other hand in colder regions pigmented varieties like Blood Red sweet oranges develop excellent quality. Warm tropical climate leads to lowest acidity, sugar contents remain almost the same in tropical and sub-tropical climates. Other contents that contribute to richness of flavor like essential oils, esters seem also more abundant in some tropical climates. All those areas and high valleys with severe and prolonged cool season should be avoided because there is risk of injury from cold.

Intercropping

Intercropping in between the trees with wheat and cotton is the common practice in Pakistan. Heavy attack of a large number of insects and pests on cotton and different requirements of irrigation to wheat damage the trees seriously. Intercropping for early income may be practiced during the initial unproductive stage of the trees. The crop selected for the purpose of intercropping must not be tall growing, very exhaustive, attractive host of insects, pests and diseases like cotton and sugarcane. Intercropping with such crops that can add humus to the soil like legumes except berseem (*Trifolium alexandrinum*) fodder is

recommended once every 3-5 years. The fodder crop berseem is most commonly sown as intercrop in Pakistan, which is not a good practice. Berseem due to its luxurious water requirements needs frequent irrigations whereas the citrus fruits are very sensitive to excessive and frequent irrigations. During early stages of tree development low moisture requiring vegetables like watermelons can be used as intercrop in citrus. However as a good principle for obtaining high yields of citrus fruit of best quality, it is recommended to manage clean cultivation. Clean cultivation also helps protect frost injury.

Weed Control

Weeds compete for irrigation water, nutrition, space, light and may also harbor pests and diseases which can affect the normal growth and fruit production. Control of weeds and grass beneath citrus trees is necessary to reduce competition for fertilizer and water. A good weed control is essential for rapid establishment and vigorous growth of young citrus trees. Weed out all grasses and weeds in the fields particularly 1-1.5 m all around the tree. Weed control can be accomplished by mechanical means or herbicide application. Mechanical means includes hoeing and cultivation. Weeding by tractor may be more effective and economical. Herbicides such as Roundup can be used to control existing weeds. Pre-emergent herbicides can also be used to prevent weed seeds from germinating.

Pest Control

Many diseases and pests attack Citrus trees and are mainly controlled by spraying and dusting with insecticides and fungicides. Citrus pests include insects, mites and diseases which may affect the health and vigor of the trees or the appearance of the fruit. Few pests actually kill citrus trees, but those that do are noteworthy because prevention is the only control.

Irrigation

The soil moisture reservoir at root level, depending on season and meteorological factors exhausts through plant use and evaporation from the soil surface. Citrus trees generally do not use much water in the cooler months of winter because weather conditions are not too favorable for

their growth and also evaporation is very low. The frequency and amount of water to be applied for irrigation depend upon the kind and age of the tree, rootstock used, stage of growth and development, climatic and weather conditions and soil type.

During summer the citrus orchards should be given restricted watering, but a heavy irrigation about two weeks before blooming has been noticed favoring increased blooming. Although the citrus trees are evergreen and needs frequent watering throughout the year but it has been observed that over-irrigation is generally more harmful for citrus fruit production than under-irrigation. It must also be kept in view that too little water does not completely replenish the soil reservoir, but excess water percolates down through the soil and is lost. Because it is not practical to replace soil moisture at the same time and rate as it is being used, watering should be scheduled periodically to refill the soil reservoir.

For practical purposes, irrigation can be applied through the basin system which is build in the shape of a watering ring atop the ground around the tree, about 10-15 cm high. The ring should be wider than the planting hole. A channel connects the basin of each tree and is very useful to conserve water. Basin system is also very useful for fertilizer application. The irrigation must be given to citrus trees when the young leaves show signs of mild wilting. In severe wilting the growth of young shoots will cease which may affect the fruit development.

Nutrition

In all about 16 elements are considered essential for getting satisfactory growth in citrus. Three major elements i.e. hydrogen, oxygen and carbon are available from water and air. The soil must provide the other 13 nutrient elements essential to all plant growth. Our soils generally are quite fertile and contain adequate quantities of all essential elements except nitrogen, phosphorus and potash (NPK). The other elements rarely need to be applied to mature, established citrus trees. However, under high pH (alkaline) soil conditions foliar spray particularly of trace elements like zinc, manganese, iron, copper, and boron has been observed to help overcome the effects of malnutrition.

Application of farm yard manure (FYM) or compost has been considered best for good growth of trees. FYM should be applied during November-December giving sufficient time to decompose before spring

the period of active growth. In case of non-availability of FYM green manuring of any leguminous crop should be done once every three years. Nitrogen is the most deficient of all elements and is needed in fairly large quantities amounting to 1-1.5 kg/tree depending upon soil condition, growth stage and variety. Mature, bearing citrus trees should receive enough nitrogen to provide for good but not excessive growth. The fertilizer may be split applied into two or three applications. The first dose in February, the second in March at the completion of fruit set, the third in August-September.

The most expedient application is to spread the fertilizer uniformly on the soil surface under the tree canopy and slightly beyond and water it in thoroughly. It is not necessary to thoroughly hoe the soil before application of fertilizer as a thorough watering will carry surface-applied fertilizer throughout the soil profile. A brief schedule of fertilizer application and manuring of citrus trees follows:

- Apply FYM at the rate of 40 to 80 kg per tree during the winter season in Dec/Jan before flowering
- Apply 3-4 kg SSP and 1.5 to 2 kg of Urea 15-20 days (Jan/Feb) before flowering
- Supply a further amount of Urea of 1.5 kg per tree after fruit setting (Mar/Apr)
- If necessary 1.5 kg of Potassium Sulphate may be applied at the time of applying FYM
- Watering is essential after each application of fertilizer.
- In case the soil pH is high, apply gypsum or any other suitable soil conditioner two to three times from blooming to fruit maturity.

Pruning and Training

Pruning is necessary for developing a strong tree skeleton and it helps prolong bearing. Training refers to systematic pruning of young trees to provide a strong skeleton and proper natural shape. For the formation of a

good framework in young trees, 2-4 evenly spaced limbs should be selected within 60-75 cm of the ground to provide a balanced and symmetrical head, and maximum strength.

Heavy pruning is generally not practiced in citrus except a light to moderate annual pruning during late winter to allow sunlight to reach inside branches. The diseased branches, dead or damaged wood, water sprouts, abnormal or undesirable growth and shoots from below the head, whether on the rootstock or the scion, should be removed as soon as they appear.

Picking

The citrus fruit should be picked only when fully ripe because citrus fruits do not ripen after picking. Lemons and limes are, however, picked green but they should be well grown to a desirable size. If left to ripen on the trees, the lemons may enfeeble their acidity, characteristics flavor and color.

In Pakistan the fruits are mostly hand picked by pulling, plucking, twisting and snapping and hand collecting in heaps, baskets, wooden or paper crates and bags. In progressive countries, for proper harvesting several types of fruit clippers, picking bags, baskets, field baskets and ladders are used. It has been observed that clipping is superior to pulling. In pulling a portion of peel is detached along with the stalk, which results in early spoilage of the fruit after harvesting.

CITRUS PROPAGATION

Propagation

The citrus trees can be propagated by seed and vegetatively by budding, grafting, cuttings and also by layering. In practice most of the commercial varieties are propagated by various vegetative methods whereas rootstocks generally are propagated through seed. All the polyembryonic varieties of citrus trees like most of seed oranges, grapefruit and tangerines will come true from seed.

To check the polyembryony, carefully peel off the outer and inner seed coat. The polyembryonic seed will have many embryos which would be convoluted upon each other. The mono-embryonic seeds with one embryo having two distinct cotyledons will not come true from seed. However in polyembryonic citrus, some of the embryos are of gametic origin and therefore do not come true. The percentage varies by species and variety.

Advantages of Seeding

The advantages of seeding include: it is much less labor intensive to simply sow citrus seeds and eliminate the grafting step; the seedling will most likely be free from viruses that sometimes get into the budwood that is used for grafting large numbers of trees; non-grafted citrus trees live longer and are more adapted to the environments.

Advantages of Vegetative Propagation

Vegetative propagation is preferred as the vegetatively propagated trees produce fruit of uniform quality; they remain relatively low-headed and bear earlier than the sexually reproduced ones. These factors result in saving in cultural operations and make harvesting of the crop and its marketing easier.

Cuttings from Sweet Lime, Citron, Lemon, or Tahiti Orange form roots easily; others such as Sweet Orange, Mandarin Orange and Grapefruit are more difficult to start. Cuttings are made during January-February and September-October. Terminal parts of branches or twigs, 20-30 cm long, cut evenly across their bases and a slanting cut at the upper

end about 4 cm above the last bud make best cutting. Remove all but 3 or 4 upper leaves. For quick and better results the cuttings particularly the round ends may be dipped in a root-inducing hormone and inserted half their length in clean sand. Treatment with Seradix A or Indole butyric acid enhances root initiation and increase success ratio. For better results the cuttings may also be callused before planting. A closed frame with bottom heat is a great help in rooting. When the cuttings have formed roots, they are transplanted in pots of suitable size and handled as seedlings.

Grafting of Citrus Trees

Graftage refers to any process of inserting a part of one plant into or onto another plant in such a way that they will unite and grow as a single unit. The scion is the part of the new combination which is inserted into the other plant and thereby produces the top of the plant, including branches, leaves and, ultimately, fruit. The stock or rootstock is the plant into which the scion is inserted and it produces the root system and lower trunk. The rootstock may be grown from seed, rooted cuttings or layers.

Grafting involves the use of a scion having two or more buds. There are numerous types of grafts including whip, cleft, bridge, in arch, stump, side, inlay bark, approach and others. Grafting is most commonly used to top-work existing trees, to change varieties, and to produce new trees. Veneer grafting and T-grafting are commonly used for vegetative propagation of Grapefruit and Kaghzi Lime which are difficult to bud because of spines in their leaf axils. They can also be propagated by air layering (*gooty*). Sweet lime can also be propagated by layering.

Budding

Budding is a method of asexual propagation in which only one bud is removed from the scion variety and is inserted by a special technique in the stock seedling. Budding involves the use of a scion with only a single bud attached to a piece of bark. It may or may not include a thin sliver of wood under the bark. Budding is the most commonly used technique for propagating new plants, but it is also used to top-work existing trees to a new variety. It is simple and easy to do, so anyone can be successful after a little practice.

There are several different methods of budding, including T, inverted T, chip (hanging), patch and others. Those most commonly used in citrus

propagation are the inverted T bud and the chip bud (hanging bud). The inverted T bud is used when the bark is slipping (easily separated from the wood), whereas the chip (hanging) bud is used at other times.

Budwood Selection

The selection of the scion is very important. The bud should be taken from a healthy and high yielding true-to-type tree preferably registered as basic or certified category. Buds should not be taken from water shoots or water sprouts and immature angular wood but from mature wood with white or grayish streaks on it. Buds should be collected from a tree or trees of the desired variety. Buds are commonly selected from the next to last growth flush (the wood behind the current flush) and from the current growth flush after it has matured and hardened. Older growth flushes can be used if the bark still is green.

Round twigs about the size of a pencil are preferred. The buds located in the axils of the leaves (where the leaf is attached to the wood) should be well developed, but still dormant.

After the budwood is cut from the tree, the undesirable wood and/or growth flush should be removed and the remaining budwood should be trimmed to lengths of 20-25 cm (8-10 inches). The leaves should be cut off leaving a stub of the petiole 3-4mm (1/8 inch) long to protect the buds.

Trimmed bud sticks should be labeled and used immediately or placed in plastic bags in a cool place. Include a moist paper towel to maintain turgidity and freshness. The label should include the variety, date of collection and source. Bud sticks are usually tied in bundles for ease of handling.

Storing Budwood

It is desirable to use budwood as soon after collection as possible, but it can be stored for several months under proper conditions. The bundled bud sticks should be sealed in a plastic bag and stored in a refrigerator. The optimum storage temperature is five degrees C (40 degrees F); it should not be allowed to go below two degrees C (35 degrees F). The vegetable drawer of the refrigerator is the best place.

Stored budwood should be checked every couple of weeks for the presence of mold or excess moisture in the bag. Budwood lightly affected

with mold should be carefully washed in cold mild soapy water, rinsed, and re-bagged in a clean bag. Excessively moist budwood can be lightly blotted on paper towels. Moldy, shriveled or darkened budwood should be discarded, as the buds probably will be dead.

When using stored budwood, it should be kept cool and moist. A good idea is to take enough budwood for a couple of hours use from storage.

Time of Budding

Budding can be done anytime there is a suitable stock on which the bark is slipping and when suitable budwood is available. The best times to do this are in late summer just before growth ceases or in the spring after growth starts.

The rootstocks can be budded any time during February-March and August-October. Shield budding with a T or an inverted T incision is the method most commonly used. Buds are inserted on the bark of the stock at height of 20-25 cm above ground level. This helps protect the scion getting infection from soil borne diseases and pathogens. Buds inserted in late summer remain dormant until the following spring. In the start of the spring the tops of the stocks may be cut off just above the point of insertion to force the buds into active growth.

To produce new plants choose rootstocks of pencil size to 2 cm (3/4 inch) diameter, either seedlings or rooted cuttings. The area to be budded should be pruned clean of thorns and twigs. The preferred budding height is 15 cm (6 inches) above ground level.

The union of scion and stock will take about 3-4 weeks to show signs of growth. When the scion bud sprouts in to a branch attaining a length of 25-30 cm, the rootstock is cut back close above the bud union. All the sprouts arising from the stocks below the scion point are broken off from time to time. Only one vigorous sprout from the scion shield bud should be allowed to grow, others if any should be removed as soon as they appear so as to form a clean stem about 90 cm to one meter long.

The shoot from the inserted bud is trained to stake to keep it straight and safe from injury. The young trees are usually cut off to force branching. They are usually left to grow for two seasons before transplanting. The budded field grown plants may be lifted for

transplanting in the dormant season with a ball of earth around the roots which are sensitive to long exposures. The container grown plants can be lifted all the year round. While transplanting the budded plant, care must be taken to avoid planting deep. The scion point should remain to a minimum of 30 cm above ground level.

As the bud grows, it will need to be staked and tied at regular intervals to prevent breakage. Remove all other buds and suckers from the rootstock as they appear.

Rootstocks

Most citrus types and varieties do not perform well on their own root system so they are commonly budded onto other citrus kinds which are called rootstocks as they are better adapted to certain soil conditions. Rootstocks are perhaps universally used for propagation of citrus trees in Pakistan and all citrus growing countries of the world. This practice has been adopted because trees produced by grafting a mature fruiting cultivar onto a rootstock have several important advantages.

- They combine good fruit traits with good root traits in the same tree. Most cultivars that produce good quality fruit do not produce their own roots with resistance to root diseases. Grafted trees also begin bearing fruit many years earlier than trees reproduced by seed.
- They accurately reproduce good fruit traits, while plants derived from seed often do not.
- A genetically separate rootstock gives the opportunity to alter tree size, productivity, and other traits through rootstock influence, while maintaining identical fruit characteristics.

For budding different varieties of citrus, suitable seedlings like Rough Lemon, Sour Orange, Sweet Orange or Trifoliate Orange are grown to furnish rootstocks. The rootstock of a budded tree includes all roots and the lower few inches of the trunk, whereas the scion is the trunk and all branches, leaves and fruit. All the side shots produced by the seedlings should be rubbed off as soon as they appear to develop a clean stem to facilitate budding.

The most common and extensively used rootstock in Pakistan is

Rough Lemon, which is well-adapted to most soil conditions where citrus fruits are grown. The other most important rootstock is the Sour Orange extensively used on somewhat moist and heavy soils in NWFP. In general major commercial varieties of citrus are propagated by shield budding on Rough Lemon (*C. jamberi*) rootstock. Some of the other varieties of citrus like Acid Lime, (*Kaghzi Nimbu*), Pummelo (Chakotra) and *Galgal* (Citron) can be propagated both by seed and budding on Rough Lemon or other suitable rootstock.

Transplanting

The citrus variety trees grow to different sizes and require different spacing for normal growth and development, therefore, they must be planted different distances apart. The normal row to row and plant to plant distances recommended for different kinds is 7-8 m depending on kind and variety. The recommended plant spacing in the square system for Grapefruits is 7-8 m; Sweet Oranges and Kino, 7-7.5 m; Tangerines, 5-6 m; Lemon and Limes, 5 m; and Kumquats, 4-5m. To figure out required number of plants for planting in an acre, divide 43,560 (the number of square feet in an acre) by the square feet of the distance used for setting the trees.

Usually Citrus trees are planted during their dormant season in winter. Before planting, the trees should be pruned to reduce the foliage and the number of branches. No more pruning will be needed until after the first growing season, when 3-4 branches should be selected to form the framework of the tree and others are removed. Add good soil to the bottom of the pits if needed to bring the trees to the right level. They shouldn't be planted any deeper than they grew. Plant firm is the rule. The soil around the tree should be well pressed to avoid cracks and crevices. The trees must be watered after planting and a small amount of fertilizer mixed with the soil is helpful. The ground right around the trees needs to be kept free of weeds. The rest of the ground can be planted with a cover crop of pulses, vegetables or other short growing leguminous crop.

In early spring, the beginning of summer and in early fall, fertilizer should be applied all around close to the tree, but the fertilizer should not come in direct contact with the trees. High doses of nitrogenous fertilizers must be avoided particularly in the fall. Moreover before winter the cover crop or inter crop should be cut or turned under as green manure and the

ground left bare for the winter.

Bud Sport

Sometimes on vegetative propagated trees mutations have been noticed in the form of bud sports. This phenomenon is very important for the fruit growers and researchers because the fruit in such cases may differ radically from the others on any grafted tree. The resultant fruit may be larger in size, superior in quality or otherwise. The foliage on such branches commonly show quite unlike characteristics found on other branches of the same tree. Such branches if noticed must be identified and the fruit carefully examined. The fruit may be a new excellent variety and the bud can provide the scion for further multiplication of the new superior variety.

CITRUS DESCRIPTOR

DESCRIPTOR	DESCRIPTION
1. Main braches: attitude	I. Upright II. Spreading III. Semi spreading IV. Drooping
2. Young shoot: anthocyanin.	I. Absent II. Weak III. Medium IV. Strong
3. Water shoot- spines	I. Absent II. Short III. Medium IV. Long
4. Leaf blade: cross section	I. Straight II. Concave
5. Leaf blade: undulation	I. absent II. Present
6. Leaf blade: firmness	I. weak II. Medium III. Strong
7. Leaf blade: tip	I. Pointed II. Rounded

DESCRIPTOR	DESCRIPTION
8. Leaf blade: base	I. Narrow II. Medium III. Broad
9. Petiole: wings	I. Absent II. Short III. Medium IV. Large
10. Anther: color	I. White II. Pale yellow III. Yellow
11. Flowering: habit	I. Once II. More than once
12. Inflorescence: type	I. Single II. Clustered
13. Fruit: size	I. Small II. Medium III. Large
14. Fruit: shape	I. Oblate II. Globose III. Ovoid IV. Obovoid V. Oblique VI. Pyriform VII. Ellipsoid VIII. Others

DESCRIPTOR	DESCRIPTION
15. Fruit: shape basal end	I. Long neck II. Short neck III. Rounded IV. Truncate V. Medium depressed VI. Deeply depressed
16. Fruit: shape of distal end	I. Depressed II. Truncate III. Rounded IV. Slightly nipped V. Prominently nipped
17. Fruit: color of surface	I. Green II. Yellowish green III. Yellow IV. Orange yellow V. Orange VI. Orange to red VII. Pink VIII. Red IX. Purple
18. Fruit: relief of surface	I. Smooth II. Ribbed III. Bumpy IV. Wrinkled

DESCRIPTOR	DESCRIPTION
19. Fruit: rind conspicuousness oil glands	I. less conspicuous II. Conspicuous III. Very conspicuous
20. Fruit: density of oil glands	I. Sparse II. Medium III. Dense
21. Fruit: style persistence	I. None II. Partial III. Total
22. Fruit: areola	I. Absent II. Present
23. Fruit: navel	I. Absent rare II. Occasional III. Always
24. Fruit: navel protrusion	I. Absent II. Weak III. Medium IV. Strong
25. Fruit: navel diameter	I. Very small II. Small III. Medium IV. Large V. Very large

DESCRIPTOR	DESCRIPTION
26. Fruit: thickness of rind	I. Thin II. Medium III. Thick
27. Fruit: rind adherence	I. Very weak II. Weak III. Medium IV. Strong V. Very strong
28. Fruit: color of albedo	I. White II. Greenish III. Pinkish IV. Reddish V. Pale green
29. Fruit: flesh color	VI. Yellow VII. Orange VIII. Pink IX. Red X. Purple
30. Distribution of red color	I. Uneven II. Even
31. Columella structure	I. Hollow II. Semi-hollow III. Solid

DESCRIPTOR	DESCRIPTION
32. Columella size	I. Small II. Medium III. Large
33. Number of segments	I. Few II. Medium III. Many
34. Fruit: juiciness	I. Weak, low II. Medium III. High
35. Fruit: juice color	I. Light green II. Yellow III. Yellow to orange IV. Red to orange V. Orange VI. Red to orange VII. Pink VIII. Red IX. Purple
36. Fruit: Juice: acid content	I. Very low II. Low III. Medium IV. High V. Very high

DESCRIPTOR	DESCRIPTION
37. Fruit: total soluble solids	I. Low II. Medium III. High
38. Fruit: strength of fiber	I. Weak II. Medium III. Strong
39. Fruit: poly-embryony	I. Absent II. Present
40. Seed: fresh color	I. Ivory II. Greenish III. Yellowish IV. Brownish
41. Seed: chalaza spot	I. Present II. Absent
42. Seed: dry color	I. Ivory II. Greenish III. Yellowish IV. Brownish
43. Color inner seed coat	I. White II. Brown III. Red
44. Color of cotyledons	I. White II. Green III. Green to white

DESCRIPTOR	DESCRIPTION
45. Time of maturity	I. Very early II. Early III. Medium IV. Late V. Very late

STATUS OF GPUS

**GERMPLASM UNITS
ESTABLISHED BY THE
PROJECT FOR HORTICULTURAL PROMOTION (PHP),
MAINTAINED BY THE
AGRICULTURAL RESEARCH DEPARTMENT
NWFP**

Background

A Pilot Project; “Malakand Fruit and Vegetable Development Project (MFVDP) was initiated in NWFP with financial and technical assistance of the Swiss Government in 1988. The project had its Headquarters at North Mingora in the beautiful valley of Swat. The project focused at development and diversification of farm production into high value crops. The project selected fruits and vegetables for improvement as they are the most important segment of the agricultural sector in NWFP. The fruits and vegetables are considered possessing potential for economic growth and capable of playing a prominent role in increasing farm profitability.

The Master Plan project included the Project for Horticultural Promotion (PHP) which was initiated in 1990. The PHP envisaged the production and marketing of certified nursery trees, raising of income of small farmers through improved marketing, better post-harvest management coupled with latest packages of production technology. It was also the objective of the Master Plan to strengthen the existing institutions in Public and Private sectors.

Project Objectives

The Project for Horticultural Promotion (PHP) aimed at achieving the following targets:

1. Integrated Crop Management (ICM): This covered better management of horticultural crops and related problems. For this purpose the ICM tried to educate the farming community through Farmers’ Field Schools and training of Trainers etc.
2. Integrated Marketing Services (IMS): The IMS covered tracing of post-harvest problems of vegetable and fruit sectors and their solutions on self-sustainable basis.

3. **Quality crop Seeds and nursery fruit Tree Production:** It was initiated “to provide a sustainable human and institutional framework for the support of small scale farmers and nurserymen, who commercially produce certified, high quality vegetable seed and fruit tree nursery plants on regular basis for the local and international markets”.

Institutional Coordination

The PHP vision has its basis on the policy that both the Public and private sectors should share the responsibilities of producing and marketing quality seed and fruit nursery trees. The public sector research institutions and the seed organizations shall limit their role to conduct basic and applied agricultural research, development of improved varieties and production of seed and propagating materials up to pre-basic and basic levels. The production and marketing of certified seed and plant materials was the responsibility of private sector. To achieve these objectives, the PHP successfully coordinated the following organizations to join hands for this purpose:

- **Research:** Agricultural Research System, NWFP, The research will develop and maintain commercial varieties of field and horticultural crops true-to-type, and produce pre-basic seed and propagating materials for further multiplication..
- **Extension:** Agricultural Extension System, NWFP, The extension service as a bridge will help take the message across so as the research in laboratory is translated in the fields in shortest possible time.
- **Private sector:** Private sector nursery men and farmers. The private sector will establish fruit plant nurseries to produce and market certified true to type, healthy and disease free fruit nursery plants. The private sector must enjoy support and interaction with research and extension.
- **Quality control:** Federal Seed Certification and Registration Department. The system will help control and regulate the production and marketing of propagating

materials following specific standards. The quality control system shall be well equipped with laboratory facilities and field expertise for indexing and monitoring of fields and markets.

Establishment of Germplasm Units (GPUs)

The varieties deteriorate or lose their identity unless maintained true-to-type and pedigreed seed or propagating materials made available for commercial production. In Kinao for example one observes a large variation in the physical characteristics of fruit like shape, fruit size, taste, rind thickness, and seediness.

To maintain and provide disease free, healthy and true to type basic material to the private sector nurserymen for the product of certified nursery plants on sustainable basis, the project has established the following three germplasm Units (GPUs):

1. **GPU Baikan** for temperate fruits at 5.3 ha at Baikan, Swat.
2. **GPU Sherkhana** for Sub-tropical fruits at about 3 ha at Sherkhana, Malakand Agency.
3. **GPU DI Khan** for Tropical fruits at 14 ha at Rakhzandani, D.I.Khan.

The germplasm entries particularly commercial varieties together with promising rootstocks representing all major fruit crops of Pakistan are maintained disease free under local environment conditions. The true to type and healthy propagation materials of research and commercial importance are available for multiplications to all research workers, registered nurseries and enlightened orchard growers of Pakistan.

Registration of Nurseries

The project also envisages regular production and supply of certified nursery plants through the strengthening of fruit tree nursery Sub-sector by arranging training for nurserymen and registration of suitable nurseries. Federal Seed Certification and Registration Department has registered a number of nurseries having good experience, well equipped with trained nurserymen and adopting prescribed standards.

Indexation

To make available true-to-type and disease free mother plants on sustainable basic, the mother stocks both of Founded and Multiplied block have to be indexed against prescribed virus diseases of each fruit species. The Federal Seed Certification and Registration Department handles this activity on regular basis according to the prescribed standards.

There is very good recognition of the efforts of the PHP as the registered nurseries are receiving orders from home and abroad and have successfully exported considerable number of nursery trees of various fruits.

The detail of the entries grown and maintained in each of the three GPUs by the Agricultural Research Department (NWFP) follows:

GERMPLASM UNIT (F) TEMPERATE, BIAKAN, MATTA, UPPER SWAT

S. #	Fruit Species	Varieties	Source
1	Apple	Royal Gala Mondale Gala Tresco Gala Gala Must Jonica Jona Gored Fiesta Golden Delicious Spartan Fuji Katy Red Falstaff Red Chief Summer Red Oregon Spur Smooth Discovery Pink Lady Sun Downer Naga Fu-2 Lady Williams Scrumptious Red Windsor Jester	Frank P Matthews Limited, UK Do Do Do Flemings Nurseries Pvt. Limited, Australia Do Frank P Matthews Limited, UK
	Tall Varieties 24		
2	Pear	Shinsui Nijissiki Hosui Kosui Beuribosc Williams Concord	Flemings Nurseries Pvt. Limited, Australia Do Frank P Matthews
	Total Varieties 14		

		Packhams Triumph Comic Jargonelle Louis Bon of Jersey Conference Doyenne du Comice Cascade	Limited, UK Do Frank P Matthews Limited, UK Do
3	Cherry Total Varieties 6	Colney Merchant Lapins Sasha Sunburst Stella	Frank P Matthews Limited, UK UK. Do Do
4	Peach Total Varieties 8	Elberta Coronet Flam Crest Florida Gold Spring Crest Benedicte Summer Lady Ryans Sun	Flemings Nurseries Pvt. Limited, Australia Do Frank P Matthews Limited, UK Do
5	Plum Total Varieties 12	Ruby Red Santa Rosa President Angelina Wilson D.Agen Valor Giant Prune Verity	Flemings Nurseries Pvt. Limited, Australia Do Frank P Matthews Limited, UK

		Seneca Marjories Violetta	
6	Apricot Total Varieties 7	Moorpark Hunter Castle Bright Westley Helena du Roussillion Bergeron Tardif de Tain	Flemings Nurseries Pvt. Limited, Australia Do Frank P Matthews Limited, UK
7	Almond Total Varieties 2	Ferrangnes Ferraduel	Frank P Matthews Limited, UK

ROOTSTOCK AVAILABLE AT GPU (F) BIAKAN

S.#	Fruit Species	Rootstock	Source
1	Apple Total R/S 6	M-25 MM-111 MM-106 M-26 M-9 M-27	Frank P Matthews Limited, UK
2	Pear Total R/S 2	Pyrus Calleryana "D6" Quince A.	Flemings Nurseries Pvt. Limited, Australia. Frank P Matthews Limited, UK
3	Cherry Total R/S 3	Colt Tabela Gasela	Frank P Matthews Limited, UK
4	Peach, Plum, Apricot Total R/S 2	Okinawa Red Leaf Nemaguard	Flemings Nurseries Pvt. Limited, Australia

GERMPLASM UNIT (F) SUBTROPICAL, SHERKHANA, MALAKAND AGENCY

S. #	Fruit Species	Varieties	Source
1	Lemon Total Varieties = 2	Eureka Frost Frost Lisbon	INRA-CIRAD, France
2	Acid Lime Total Varieties = 4	Deperse Mexicaine Tahiti Kaghzi Lime	INRA-CIRAD, France Horticultural Research Station, Sahiwal
3	Tangerine Total Varieties = 3	Dancy Fair Child Fremont	INRA-CIRAD, France
4	Mandarin Total Varieties = 2	Kinnow Futrell's Early	Horticultural Research Institute, Faisalabad.
5	Tangelo Total Varieties = 1	Pearl	Horticultural Research Station, Sahiwal
6	Sweet Orange Total Varieties = 25	Salustiana Tarocco Rosso Jaffa Blood Red Musambi Sanguinello Marrs Early Valencia Late Succari Parson Brown Moro blood orange Sanguinelli	INRA-CIRAD, France Horticultural Research Institute, Faisalabad. Horticultural Research Station, Sahiwal CCPP, University of California, USA.

		blood orange Ruby blood orange Pineapple orange Shamouti orange Marrs orange Atwood orange Skaggs bonanza navel Parent Washington navel Fisher navel Lane late navel Spring navel Campbell valencia Olinda valencia Frost Valencia	
7	Grape Fruit Total Varieties = 6	Pink Ruby Ray Ruby Red Blush Shamber Star Ruby Ruby Red	INRA-CIRAD, France Horticultural Research Station, Sahiwal
8	Litchi Total Varieties = 8	Bedana Gola Surahi Shoi Thung	Horticultural Research Institute, Faisalabad. Fruit Nursery Farm, AED, Haripur

		(C) Purbi Bombay Mangalbari China-3	FTIP- BAU –Dhaka, Bangladesh
9	Loquat Total varieties = 2	Seedless (Imported) Haripur Local	Mr. Habibullah Khan, Charaga, Mardan, Ex Director, ADBP Fruit Nursery Farm, AED, Haripur
10	Guava Total Varieties = 11	Ramzani Sindhi Gulabi Riali Thada Rami KAZI PG-001 PG-005 PG-010 PG-013 Guava seedless	Quaid-e-Awam Research Institute Larkhana, Sindh FTIP- BAU –Dhaka, Bangladesh ARI-Tarnab, Peshawar.
11	Mango Total Varieties	Langra Chaunsa Sindhri Dosehri Anwar Retaul Sensation	Horticultural Research Institute, Faisalabad

CITRUS ROOTSTOCK AVAILABLE AT GPU (F) SHERKHANA

S. #	Fruit Species	Rootstock	Source
1	Citrus Total R/S = 7	Troyer Citrange Citrus Volkamarriana Carrizo Citrange Bigarade Australian Rangpur Lime Flying Dragon Cleopatra Mandarine	INRA-CIRAD, France

GERMPLASM UNIT (F) TROPICAL, RAKHZANDANI, D.I. KHAN

S. #	Fruit Species	Varieties	Source
1	Date Palm	Dhakki	Agricultural Research Institute, D.I. Khan
		Basra	
		Gulistan	
		Haleeni	
		Hussaini	
		Begum Jhangi	Date Research Centre, Turbat, Balochistan
		Aabidandan	
		Kuzan Abad	
		Kahraba	
		Mozawati	
		Rabai	Agriculture Extension Farm, Panjgur, Balochistan.
		Burni	
		Sabzo	
		Jawansore	
		Aseel	
		Karblain	Date Palm Research Station, Kot Digi, Sindh
		Pathri	
		Halavi	
		Makran	
		Shamran	
		Khudravi	Date palm Research Center, Jhang
		Deglet Noor	
		Zaidi	
	Total Varieties = 23		

2	Mango Total Varieties = 14	Saleh Bhai Saroli Bengan Pali Sawarnarika Alfanso Sindhri Dosehri Chounsa Anwar Retaul Yakta Chounsa Sufaid Late Retaul Sensation Langra	Sindh Horticultural Research Institute Mirpur Khas, Sindh Mango Research Station Shuja Abad, Multan, Punjab. Horticultural Research Institute, Faisalabad.
3	Guava Total Varieties = 11	Ramzani Sindhi Gulabi Riali Thada Rami KAZI PG-001 PG-005 PG-010 PG-013 Guava seedless	Quaid-e-Awam Research Institute Larkhana, Sindh FTIP- BAU – Dhaka, Bangladesh ARI-Tarnab,

			Peshawar.
4	Ber Total Varieties = 11	Sufi Sangarh White Gola Late Gola Chambalee Gooba Sufi Sialkoti Ajmairi Gilli Delhi White Mahmood Wali Karella Sufan Goar	Ber Research Institute Tando Jam, Sindh Horticultural Research Station, Bahawalpur
5	Falsa Total Varieties = 1	Falsa Large	Sindh Horticultural Research Institute Mirpur Khas, Sindh
6	Litchi Total Varieties = 4	Bedana Gola Surahi Shoi Thung (C)	Horticultural Research Institute, Faisalabad Fruit Nursery Farm, AED, Haripur

(DRAFT)
CERTIFICATION STANDARD -
CITRUS NURSERY TREES

1. Introduction

The purpose of certification program is to ensure the sanitary status and true-to-type character, along with horticultural value of the propagating material in commercial propagation. Certification also controls the registered set up of nurseries. The certification program only ensures the absence of prescribed pathogens. The requirements and certification standards follow:

2. Nursery Location and Land

Any applicant owner of a nursery shall be eligible to produce and/or market citrus nursery trees subject to fulfilling the following requirements:

- i) The nursery location shall be in an area with minimum risk of infection, preferably an area free from citrus canker, CTV, root rot, stubborn and root weevil as prescribed by the Federal Seed Certification and Registration Department. The nursery shall be away at least 15 meter from thoroughfares, roads, and croplands.
- ii) The land must be free from nematodes, and other pathogens as prescribed by the Federal Seed Certification and registration Department from time to time. An area of 100 feet around the nursery must also be kept free of prescribes nematodes and other pathogens. The nursery must have an effective nematode and *phytophthora* control program.
- iii) The nursery area must have a suitable drainage system to lead run-off irrigation or rain water out of the nursery area and also the drainage system must ensure that run-off or flood water from outside particularly from other citrus plantings in the vicinity of the nursery is prevented from entering the nursery area.

- iv) All rootstock can be grown under field conditions, on raised beds, benches preferably in polythene bags and other suitable containers.
- v) The soil, all growing media, manure, potting mix or mulch etc. added to a nursery field must be free of prescribed nematodes and pathogens.
- vi) The containers to produce citrus nursery trees must be stored on concrete or polythene ensuring that the containers do not come into direct contact with uncertified soil or growing media.
- vii) Any field site will not be eligible for citrus nursery tree production, which had previously been planted with citrus for two to three years.
- viii) All nursery use implements like pruning shears and saw etc. shall be sterilized and disinfected prior to any fruit picking, bud collection, pruning or cutting on any registered tree in any block.
- ix) Each registered nursery shall maintain a detailed and accurate record of all agricultural events in the nursery particularly the field history, history of each registered tree with full record of budwood collection and fruit production, number of rootstock trees grown and certified nursery trees produced and sold.
- x) The citrus nursery site shall be subject to recommendation of the Nursery Registration Committee (NRC) and approval by the Federal Seed Certification and registration Department.
- xi) The site shall be inspected annually by the Nursery Registration Committee and plants shall be sampled routinely to verify the phyto-sanitary conditions of the nursery and freedom from prescribed pathogens.
- xii) In case the requirements have not been met, the phyto-sanitary conditions are poor or substandard or the plants have been found to be infected with prescribed nematodes or other pathogens or have knowingly sold such plants,

misrepresenting the nursery stocks as certified, the registration of the nursery shall be cancelled.

3. Establishment of Scion Source Blocks

The citrus certification program of Pakistan shall base on the following three categories of scion source blocks:

- i. Pre-basic Block (Foundation Block)
- ii. Basic Block (Multiplication Block)
- iii. Certified Block (Field Increase Block)

A. Pre-basic Block: (Primary Foundation Block)

The Pre-basic Block (Foundation Block) shall consist of field grown or container grown registered trees, which will serve as the primary source of propagating material of the variety. Major requirements follow:

- i) The Pre-basic Block (Foundation Block) shall be maintained by the Horticultural Research Institutes, Universities or any other organization set up for this purpose. [The GPU, Sherkhana, can serve as pre-basic (foundation) block for citrus varieties provided the registered mother trees are disease free and indexed regularly].
- ii) In case the mother trees are infected with vector transmitted diseases, the healthy plants shall be regenerated through micro-grafting technique or shoot tip grafting *in vitro* from infected bud lines of the GPU.
- iii) To avoid contamination with vector transmitted diseases, the regenerated trees shall be grown in field or in containers under insect-proof screenhouses or greenhouses. The protected block shall be located not less than 30 meter distant from any commercial orchard of the same genus. A distance of about 6 meters all around the block shall be kept open, weed free or clean cultivated. This block shall also be a separate planting sufficiently apart from any other block or planning to maintain its identity and sanitation.
- iv) The block shall contain true-to-type, normal growing healthy and virus free plants possessing high horticultural

qualities. The plants may have been recovered through clean stock, tissue culture (shoot tip grafting), or any other standardized technique.

- v) The trees in this block will be the primary source of budwood for the establishment of the Basic block (Multiplication Block).
- vi) For each accession 2-3 plants shall be maintained under best management conditions, pruned if necessary and should produce fruit regularly.
- vii) The plants shall be adequately spaced from each other so that their branches do not touch each other.
- viii) These trees shall be described, registered and labeled individually. The Registration number of each tree shall be clearly marked on the bark with a weatherproof marker.
- ix) These trees shall be inspected and indexed annually to examine their health status.
- x) The plants shall be examined periodically by the specialists particularly on fruiting and flushing stages to detect any growth or fruit abnormality.
- xi) Any plant showing disease symptoms or behaving abnormally shall be removed immediately.
- xii) To verify field performance a group of at least 2-3 trees originating from the same parent tree and approved rootstock shall be planted outdoors in the adaptability area of the variety for assessment of growth habit, trueness to variety, horticultural qualities and disease response.

B. Basic Block (Multiplication Block)

- i) Basic (Multiplication) Block means field or container grown plantings propagated with budwood from pre-basic block to serve as source for further propagation as certified block. The requirements follow: Basic block shall be maintained by the public sector Horticultural Research Institutes and/or registered nurseries.

- ii) The trees in this bloc shall be field or container plantings propagated with budwood from pre-basic block. The basic block shall be established in screenhouse or greenhouse to protect trees from vectors of CTV and other vector borne diseases.
- iii) The trees will be indexed annually for vector-transmissible pathogens and every three years for mechanically transmitted or other pathogens. Any infected tree will be removed immediately.
- iv) These trees shall be examined annually by the horticultural specialists for identification of possible off-type branches or any growth or fruit abnormality that could be removed.
- v) The number of trees to be maintained for each accession shall be 2-10 trees (subject to commercial requirement), each one identified by a registration number clearly marked on the bark with weatherproof marker.
- vi) The trees shall be adequately spaced from each other so that their branches do not touch each other.
- vii) To verify field performance a group of at least 2-5 trees originating from the same parent tree and approved rootstock shall be planted for assessment of growth habit, trueness to variety, horticultural qualities and disease response.

C. Certified Block (Field Increase Block)

Certified (Field Increase) Block shall mean a planting of citrus nursery stock of a variety used as scion source for producing certified nursery trees. The requirements follow:

- i) The Certified Block shall be maintained by a registered nursery.
- ii) The Certified Block shall contain field grown plants propagated directly from registered trees of basic block or pre-basic bloc.

- iii) The certified block can be established under normal field conditions or in screenhouse or greenhouse conditions to protect them from vector borne pathogens.
- iv) The Certified Block Trees shall also be registered and identified with a registration number, which shall be clearly marked on the bark with weatherproof marker.
- v) The number of trees in a Certified Block for each accession can be 4-10 or more depending upon the availability of nursery facilities and field demand.
- vi) Separate blocks shall be maintained for each accession or citrus variety.
- vii) Certified Blocks shall be inspected by Horticultural Specialists annually to detect any growth and development abnormalities and disease symptoms.

4 Seed Source Block

The Seed Source Block shall mean the registered trees to produce seed as a result of sexual recombination, which is used for the production of rootstock for citrus propagation and shall meet the following criteria:

- i) Any person, nurseryman or organization desirous of registration of a citrus tree of a particular variety as a seed source tree shall apply to the Federal Seed Certification and Registration Department declaring that the tree is a nucellar tree from a certified source or has been vegetatively propagated from a certified scion tree.
- ii) The Nursery Registration Committee (NRC) will cause a visual inspection of the perspective tree and the surrounding trees during the fruiting stage to determine that:
 - a. The tree is healthy and free from bark and leaf symptoms of prescribed complex of viruses and recognizable symptoms of other citrus diseases.
 - b. The tree is true-to-type and the morphological characteristics of the tree are normal.

- c. The tree is high yielding and the fruit quality is horticulturally true-to-type.
 - d. The tree is free of unacceptable fruit mutations, disorders and abnormal growth.
- iii) If on the inspection the perspective source tree is found to be satisfactory, the committee shall collect bud samples from the tree and arrange indexing for the prescribed complex of viruses and pathogens and if it is found to be unsatisfactory the applicant shall be informed of the decision.
- iv) If the bud samples collected from the perspective source tree are found to be negative and there is no evidence of bud mutation in the foliage, the Federal Seed Certification and registration Department shall register the tree and will allot a registration number which shall be clearly marked on the bark with a weather proof marker.
- v) The Federal Seed Certification and registration Department shall carryout annual inspection of the source tree and shall cancel in consultation with the Nursery Registration Committee the registration of the source tree if an annual inspection during fruiting reveals evidence of:
 - a. Symptoms of virus or other bud-transmittable diseases on the seed source tree or any of the trees immediately surrounding that tree;
 - b. High degree of mutation on the fruit or foliage of the source tree;
 - c. Improper keeping of records of the source tree which are likely to confuse the identity of the source tree;
 - d. Careless handling of seeds collected from the seed source tree (mixing or poor storage etc.).

5. Inspection and Indexation

To assess the health status of the propagating materials in all nursery blocks the Federal Seed Certification and Registration Department in

consultation with the NRC shall cause an examination of the stocks and take samples for indexation: the procedure follow:

4. All registered trees in the pre-basic, basic and certified blocks shall be inspected and indexed annually for freedom from nematodes, vector-transmitted diseases and other pathogens.
5. Rootstocks shall be inspected twice: firstly before grafting for purity of the stock and general health conditions and secondly two weeks after grafting to check the procedure and health.
6. In all nurseries, the certified nursery plants shall be routinely inspected and sampled by the Federal Seed Certification and Registration Department to verify that plants are free from prescribed pathogens and vector transmitted diseases.
7. Any plant found to be off-type on any inspection or showing excessive mutation, abnormal or unhealthy growth shall be removed immediately from any block.
8. The collection of the budwood from pre-basic, basic and certified blocks shall be witnessed by the representative of the Federal Seed Certification and Registration Department and proper records will be maintained.
9. Budded nursery trees shall be kept in separate rows providing sufficient gaps between rows from different budwood sources.
10. Each row shall be clearly labeled with the bud source; rootstock used and date of budding.

6. Production of Certified Nursery Trees

1. The certified citrus nursery trees shall be produced using the budwood collected from registered trees in the following manners: Certified nursery trees shall be produced only by the nurseries registered under Pakistan Fruit Nursery Rules, 1998.
2. Certified citrus nursery trees shall be produced using the budwood collected from registered trees on specified rootstock seedlings that have not previously been budded.

3. In case the re-budding is necessary, budwood from the same bud source as the original shall be used.
4. All budwood collection from registered trees of any block shall be witnessed by the representative of the Federal Seed Certification and Registration Department.
5. The budwood shall be collected in budwood bundles which shall carry a certificate mentioning the variety, nursery registration number, tree registration number, number of buds collected and date of collection.
6. Every registered nursery shall maintain a detailed and accurate record of history of each registered tree, collection of budwood, budding and re-budding and the movement of the certified nursery trees.

7. Registration of Source Trees

All trees in pre-basic, basic and certified blocks and approved rootstock shall be registered on the basis of the following criteria:

1. **Genuineness:** The tree must be true-to variety description as established jointly by the Federal Seed Certification and Registration Department and horticultural specialists of the maintaining organization.
2. **Health status:** The tree must be healthy and free from diseases and pathogens.
3. **Horticultural value:** The tree must be high yielding and possessing high horticultural quality, which shall be assessed jointly by the Federal Seed Certification and Registration Department and the Horticultural specialists on the basis of at least two years field performance.

8. Labeling

- I. All the nursery trees offered for sale or held in stock for sale or distribution at any nursery or sale point shall bear a tag according to the following order: Pre-basic nursery stocks White with violet diagonal line
- II. Basic nursery stocks White
- III. Certified nursery trees Blue

IV. Approval nursery trees Pink

V. Truthfully labeled trees Yellow

A. Certified Nursery Trees

Each of the certified nursery trees shall be identified by a Blue tag bearing the following information:

- I. Name and Registration Number of the Nursery
- II. Kind of fruit
- III. Name of variety
- IV. Date of budding
- V. Source of budwood (registration number of the source tree)
- VI. Rootstock variety
- VII. Category Certified/Approved

B. Truthfully Labeled Nursery Trees

Any citrus nursery offering nursery plants for sale except certified nursery trees shall have to label such nursery trees under Seed (Truth in Labeling) Rules, 1991 with a yellow tag bearing the following information:

- I. Name and registration number of the nursery
- II. Name of kind and variety
- III. Date of budding
- IV. Source of budwood
- V. Rootstock
- VI. Category Truthfully labeled

NOTES AND TERMS

1. **Acuminate:** The shape of a tip or base of a leaf or perianth segment where the part tapers gradually and usually in a concave manner.
2. **Adventitious:** Produced in an unusual position or time; can refer to any plant organ (bud, root, shoot).
3. **Albedo:** A mesocarp of a citrus fruit, the white spongy tissues lying between the colored peel and the juice sacs. It is full of vitamins.
4. **Anthocyanin:** A class of color soluble pigments responsible for the red, purple and blue coloration of flowers, fruits, and other plant parts.
5. **Apex:**
6. **Areola:**
7. **Bare-root:** Bare-root plants are offered for sale with the soil removed from around the roots. These dormant plants are dug from growing fields, trimmed and freed of soil, and then protected against drying out until planting.
8. **Base:**
9. **Beneficial organisms:** These are the much favored insects, arachnids, nematodes, and other organisms that eat or parasitize harmful insects and mites. Many beneficials already exist in our gardens, others can be introduced.
10. **BT (*Bacillus thuringiensis*):** A bacteria introduced in plant tissues to kill caterpillars attacking the crop plants.
11. **Bud:** Little swollen areas along a stem, branch, or trunk where new growth may occur. Cutting above one of those buds stimulates growth.
12. **Bud sport:** Sometimes on vegetative propagated trees mutations have been noticed in the form of bud sports. This phenomenon is very important for the fruit growers and researchers because the fruit in such cases may differ radically from the others on any

grafted tree. The resultant fruit may be larger in size, superior in quality or otherwise. The foliage on such branches commonly show quite unlike characteristics found on other branches of the same tree. Such branches if noticed must be identified and the fruit carefully examined. The fruit may be a new excellent variety and the bud can provide the scion for further multiplication of the new superior variety. This phenomenon has been noticed more pronounced in citrus.

13. **Bud union; graft union:** A swollen area just above the soil level where one variety has been grafted onto the rootstock of another variety. The bud union is not always swollen, and on some older plants it can be difficult to find. On young bare-root trees the bud union may look like a slight bend.
14. **Cambium:** A thin formative layer between the xylem and phloem of most vascular plants that gives rise to new cells and is responsible for secondary growth.
15. **Cane:** The woody stem of a rose or a berry plant, it can also be the jointed and often hollow or pith-filled stem of a bamboo or sugar cane.
16. **Chance seedling:** A term commonly used in reference to a cultivar's origin. It is a seedling arising by "chance" (from random, naturally occurring pollination), selected for desirable horticultural traits, and propagated vegetatively to maintain it as a cultivar or clone. Most of the popular and economically important fruit varieties have been selected as chance seedlings.
17. **Chilling requirement:** The number of hours of temperatures required below 45°, and refers most often to fruit trees that need a certain amount of cold weather in winter in order to bloom and bear fruit well the following year.
18. **Clonal rootstocks:** Clonal rootstocks are vegetatively propagated. Degree of size control and anchorage varies among dwarfing rootstocks. Choices outside of apple for size controlling rootstocks are more limited.
19. **Clone:** A clone is a genetically identical group of plants derived and maintained from one individual by vegetative propagation.

20. **Cold hardiness** (hardy): The ability of plants to withstand cold injury (*autumn-winter*).
21. **Colomella**:
22. **Cordate**: Heart-shaped, often restricted to the basal portion rather to the outline of the entire organ.
23. **Cross** pollination: The transfer of pollen from the anther of the flower of one plant to the flowers of a different plant.
24. **Cultivar**: A cultivar is a contraction of the words "cultivated" and "variety." It is a plant raised or selected in cultivation that retains distinct, uniform characteristics when propagated by appropriate methods. [A race or variety of a plant that has been created or selected intentionally and maintained through cultivation].
25. **Cutting**: A section of a plant that is cut off and rooted to create a new plant.
26. **Deciduous**: Deciduous plants drop all their leaves once a year.
27. **Dioecious**: Having male (staminate) and female (pistillate) flowers on different plants.
28. **Dormant**: The term dormant describes the inactive or sleeping state in which a plant stops growing but is still alive.
29. **Dwarfing Rootstocks**: These restrict growth and tree size for reasons such as slower nutrient uptake or smaller root systems. As a rule, these are asexually propagated, but can occur from seed (*trifoliate orange*).
30. **Elliptical**: An object oval in outline with rounded ends and widest about the middle (elliptic).
31. **Frost Damage**: Low temperature injury during some stage of the growing season. Parts affected are flower buds, flowers, and young fruit (*spring*) or near mature fruit or other tissues (*fall*).
32. **Genetic Dwarf**: Genetic dwarfs are trees in which the internodes of the cultivar (*scion*) are greatly compressed. They are naturally small (*five to eight feet*) trees, even when grafted or budded onto standard, non-dwarfing rootstocks.
33. **GMOs**: Genetically modifies organisms.

34. **GPU:** The germplasm units of tropical and sub-tropical fruit trees, established by the project for horticultural promotion (PHP), in NWFP.
35. **Grafting:** A way to propagate a plant by inserting a section of one plant (the scion) into another plant (the stock, rootstock).
36. **Hardening:** The process of achieving cold hardiness. In a broad sense it's the process that increases plants ability to survive the impact of unfavorable environmental stress. Water, nitrogen, other practices can affect this process
37. **Heading Back:** A type of pruning cut by which the end of a branch is removed to stimulate branching farther back on the branch or trunk.
38. **Heel In:** A means of preventing roots of bare-root plants from drying out before they can be set out in the garden. Dig a shallow trench; lay the plant on its side so that roots are in the trench, then cover roots with soil, sawdust, or other material, moistened to keep roots damp.
39. **Honeydew:** A sticky substance secreted by aphids and several other sucking insects.
40. **Integrated pest management (IPM):** An environmentally sensitive system of pest control in which beneficial insects are used and little or no chemical spraying is needed. When insecticides are used, they are targeted at a specific pest rather than a broad spectrum of insects which may also include beneficials.
41. **Lanceolate:** A lance-shaped, much longer than broad being widest at the base and tapering to the apex.
42. **Maintainer:** The research institute, developing organization, the breeder, or any person authorized to maintain a crop variety (genotype) true to registration and make available true to type seed and propagating materials for further multiplication on regular or cyclic basis.
43. **Monoecious:** Separate female and male flowers born on one plant.
44. **Nucellus:** The nutritive tissue in an ovule.
45. **Nucellar Embryo:** The embryo developed from the tissue of the

nucellus and consequently of the same genotype as the female parent.

46. **Oilglands:**

47. **Ovoid:** A solid object which is egg-shaped in section.

48. **Partially Self-ferile:** A heavier fruit crop is set when two or more varieties are planted nearby.

49. **Petiole:** The stalk of a leaf.

50. **Pinnate:** A compound leaf consisting of several leaflets arranged on each side of a common petiole.

51. **Pollination:** The transfer of pollen from the male part of flowers (the anthers) to the female part (a stigma). The transfer is accomplished by insects or other means.

52. **Polyembryonic:** The production of two or more embryos within an ovule. In the state of polyembryony, generally, one of the embryos in the seed is a hybrid (zygotic); whereas the others are vegetative in nature. These vegetative growths can faithfully reproduce the true characteristics of the parent variety. In citrus the seed is polyembryonic containing both sexual and asexual seedling. Therefore, while propagating by seed, the sexual seedlings which are usually stunted and poor are rogued out. The rest that are produced from the cells of the nucellus are more vigorous and allowed to grow. The seedlings, thus selected, are more or less uniform in growth and production.

53. **Rachis:** The main stalk of a flower cluster or the main leafstalk of a compound leaf.

54. **Rind:** The protective outer layer or skin on fruits.

55. **Rootbound:** When a plant roots become tangled, matted, and grow in circles. Always untangle rootbound plants when replanting them in a larger container or in the ground.

56. **Rootstock:** Sometimes called "stock," this is the root system (*plant*) propagated from seed (*seedling*) or vegetatively as common in clonal rootstocks on which various cultivars is budded or grafted. Many rootstocks are used and possess traits that relate to anchorage, size control, tolerance of light and heavy soils, "wet

feet", specific nematodes and other plants and diseases.

57. **Scion** (*pronounced si-on*): A detached stem, usually dormant, used in asexual propagation by grafting techniques.
58. **Seedling Rootstocks**: Seedling rootstocks are propagated from seed selected from various cultivars and typically produce well anchored trees. The terms standard or standard root or standard rootstock imply that the rootstock is seedling and that the cultivar grown on that rootstock will be a full size tree.
59. **Self-fertile**: One variety is needed for pollination.
60. **Self-sterile**: Two varieties are needed for pollination.
61. **Sport**: A sport is a mutation (*genetic change*), naturally occurring or induced to arise from a bud. It differs from the parent plant and can be vegetatively propagated to give rise to a new cultivar.
62. **Spur type**: Spur type is used in reference to the cultivar's habit of growth; on a given amount of wood, spur type cultivars produce more fruiting spurs than do non-spur type cultivars.
63. **Spur**: A spur is a short shoot with compressed internodes. They grow from two year or older branches and produce flowers and fruit. Flower spurs are the best exemplified in apple and pear trees.
64. **Stigma**: That part of a pistil through which fertilization by the pollen is effected.
65. **Sucker**: A cane that emerges from below the bud union, and therefore comes from the rootstock rather than from the variety grafted onto it. On other plants, a sucker is any unwanted, fast-growing, upright growth from roots, trunk, crown, or main branches.
66. **TSS**: Total soluble sugars.
67. **Variety**: Variety and "named variety" are commonly used to mean the same as cultivar. Technically, it means a naturally occurring variant of a species.
68. **Vesicles**: A small sac or cavity filled with fluid or air.
69. **Winter burn**: Associated with desiccation. Certain atmospheric conditions (*wind*) along with frozen ground and/or reduced soil

moisture may not meet transpirational demands - a particular problem with evergreens.

70. **Winter injury:** Low temperature injury during rest (dormancy) and affects buds, stems, main limbs, and trunk.

Citrus varieties

GPU, Nursery Tree Production, Certification

REFERENCES